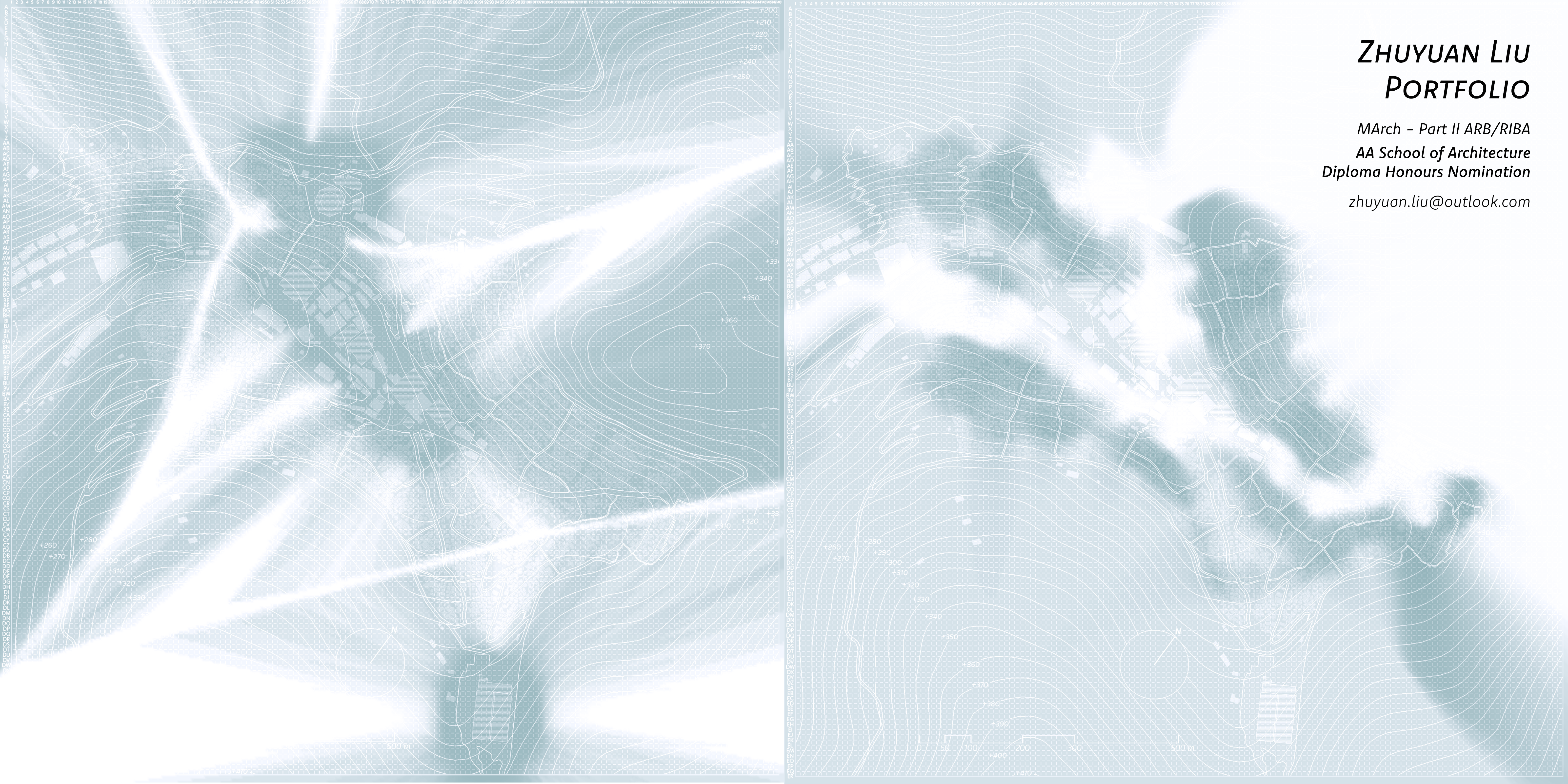




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PORTFOLIO

MArch - Part II ARB/RIBA
AA School of Architecture
Diploma Honours Nomination
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1. PROFESSIONAL PRACTICE

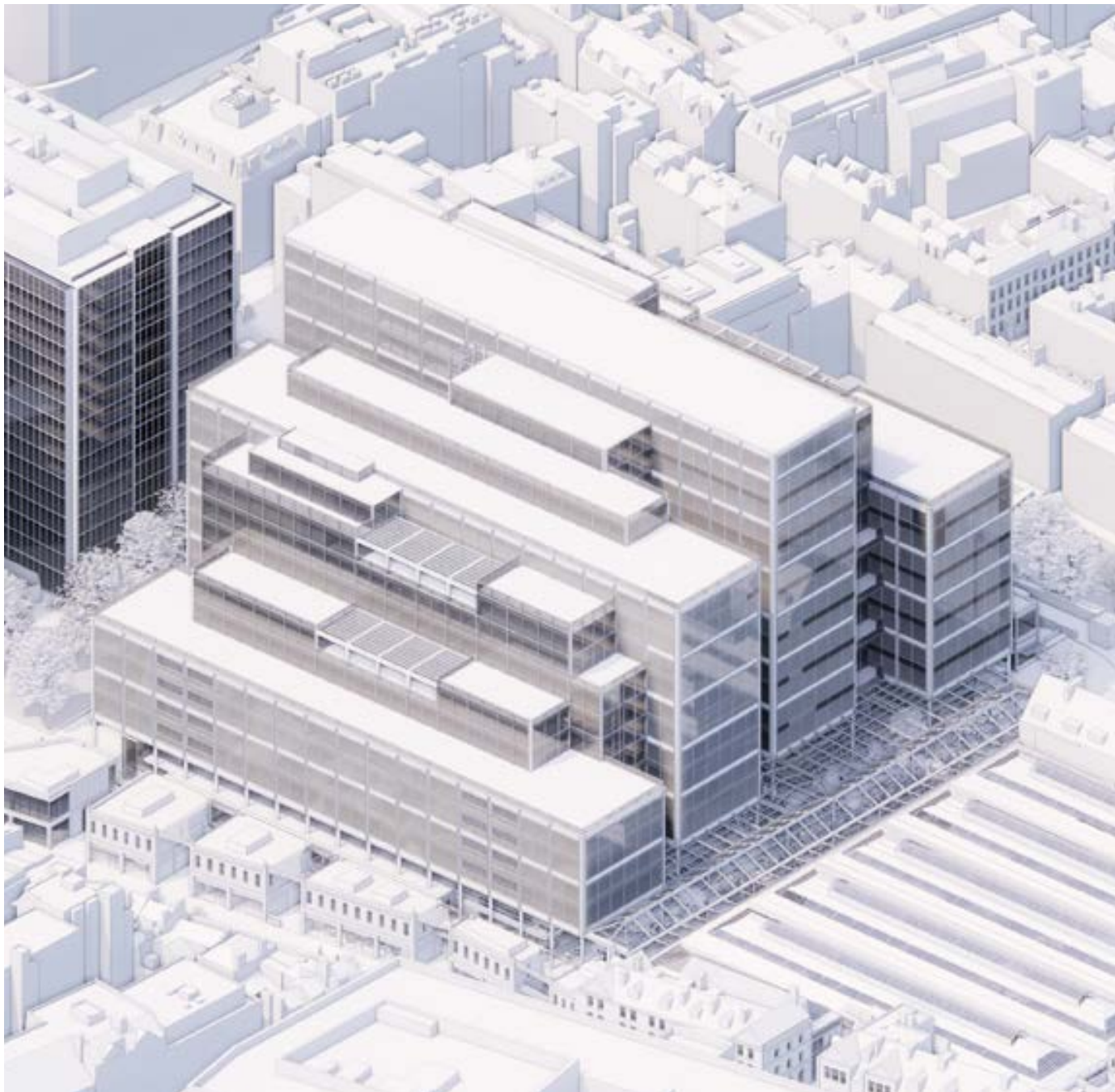
WORKING EXPERIENCE AT FOSTER+PARTNERS,
STUDIO 1

1 SPITALFIELDS

CAD, REVIT, RHINO, ENSCAPE



Market – Proposed View I



Main Building in Context



Main Building — Proposed Perspective



Spitalfields Market + East Canopy — Proposed

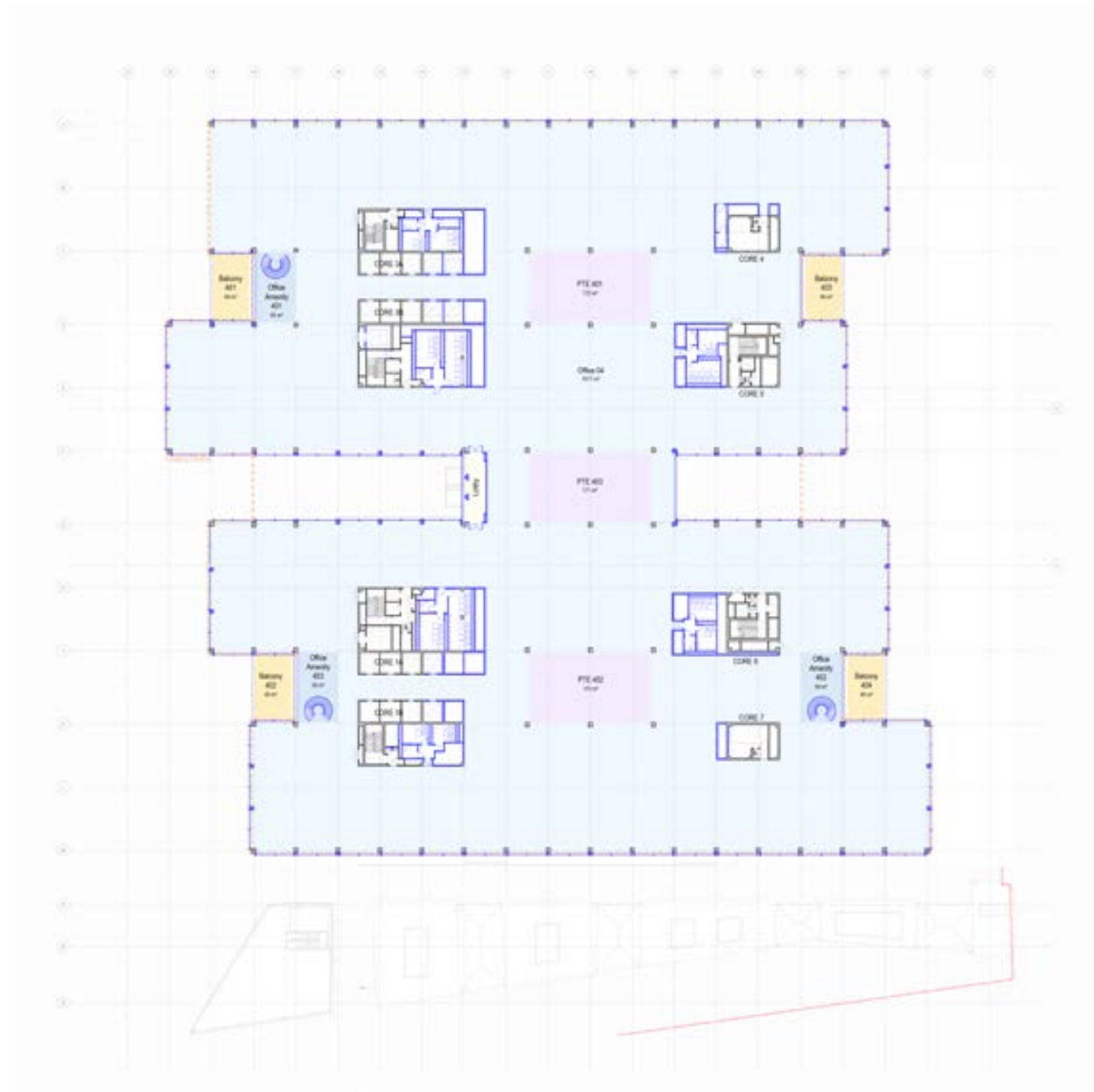
1 Spitalfields is a major refurbishment proposal connecting the main building with Brushfield Street and Spitalfields Market. The design responds to a historic urban setting shaped by the market, the church and Liverpool Street Station.

Working within the Foster + Partners Studio 1 team, I contributed to façade refinement, central atrium studies, landscape design and market planning, alongside the development of models and visualisations. These studies supported the evolving proposal and its subsequent development through coordinated Revit models and drawings.

Revit Documentation & Technical Coordination

Working collaboratively within the project team, I contributed to the production and coordination of all plans and sections shown here, covering the East Canopy connection and the wider building package.

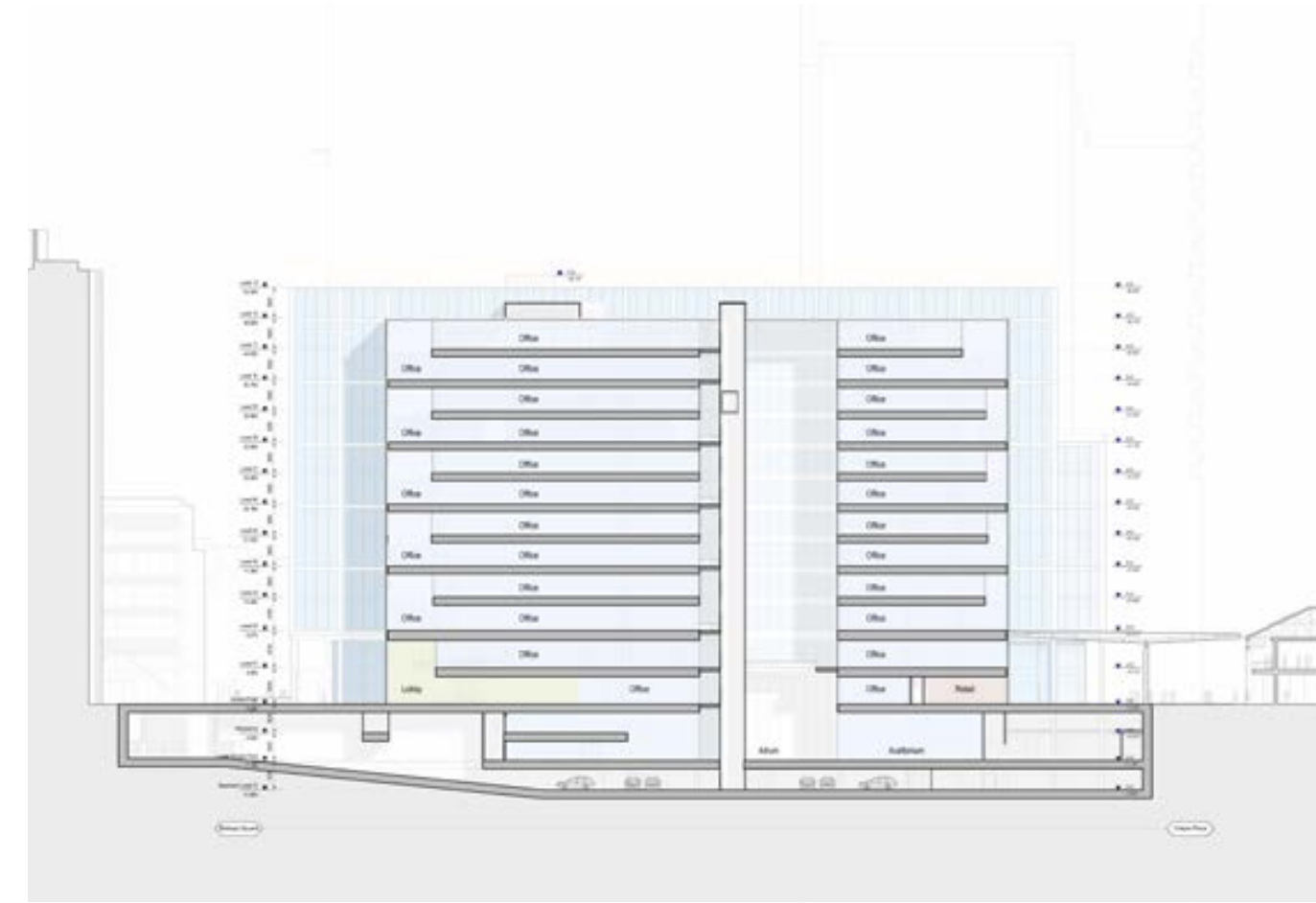
The drawings document existing conditions, proposed demolition and new interventions. My work included developing connections along the building's eastern edge and coordinating spatial, structural and circulation requirements across the Revit model and associated drawing packages.



Typical Floor Plan



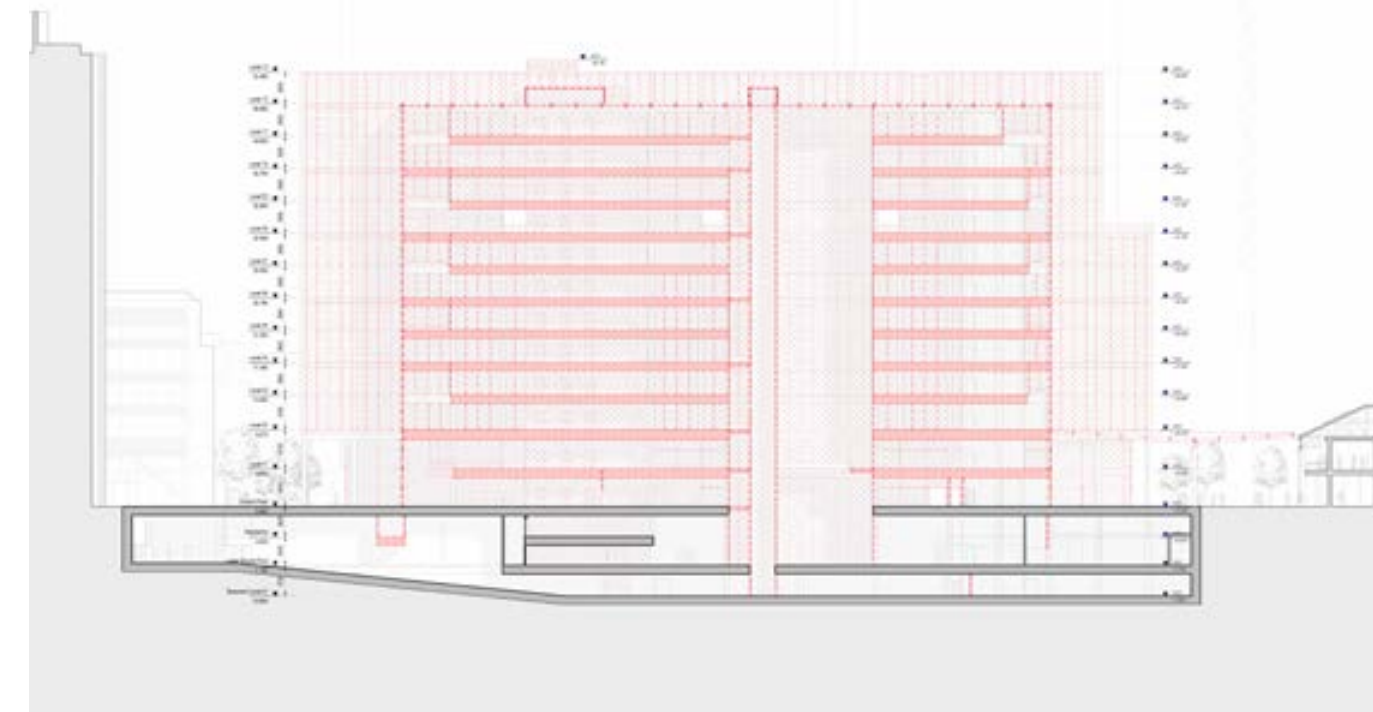
Podium Plan



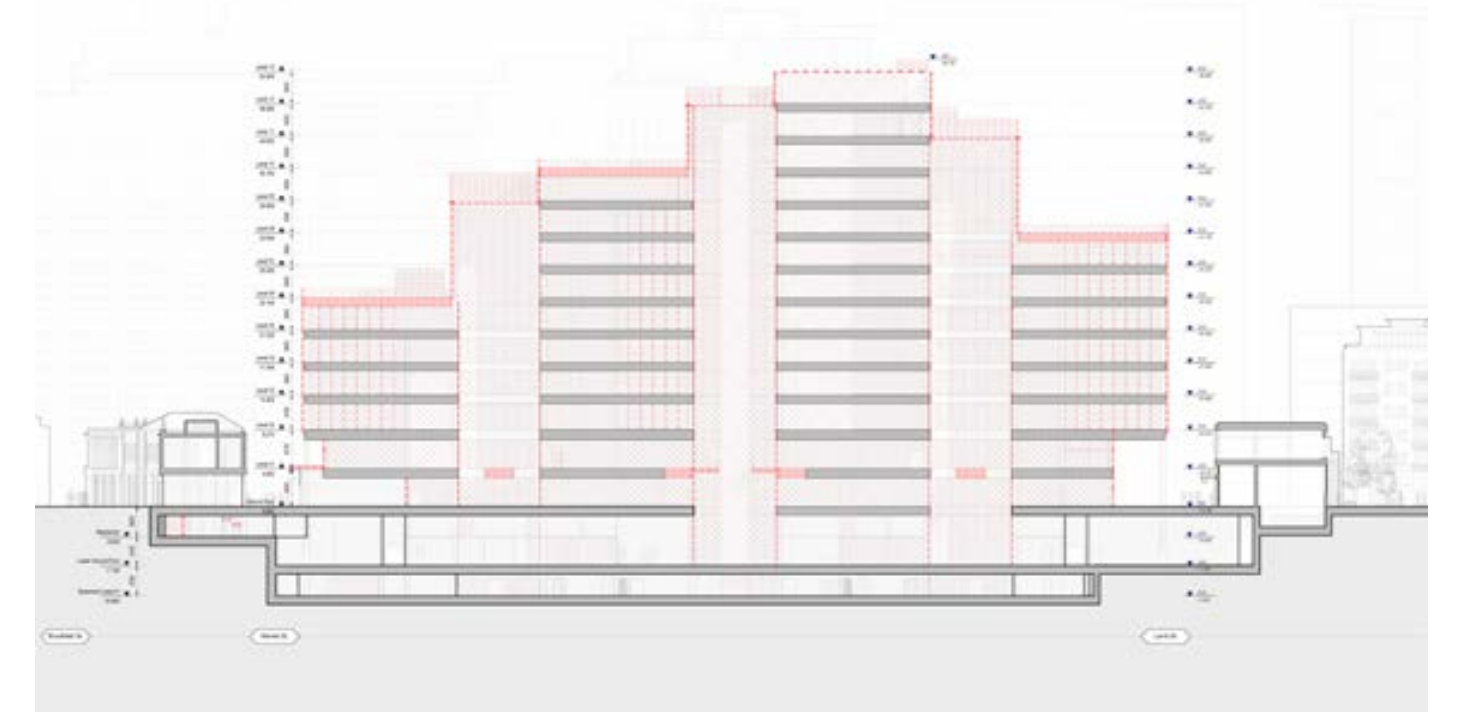
East–West Section — Existing



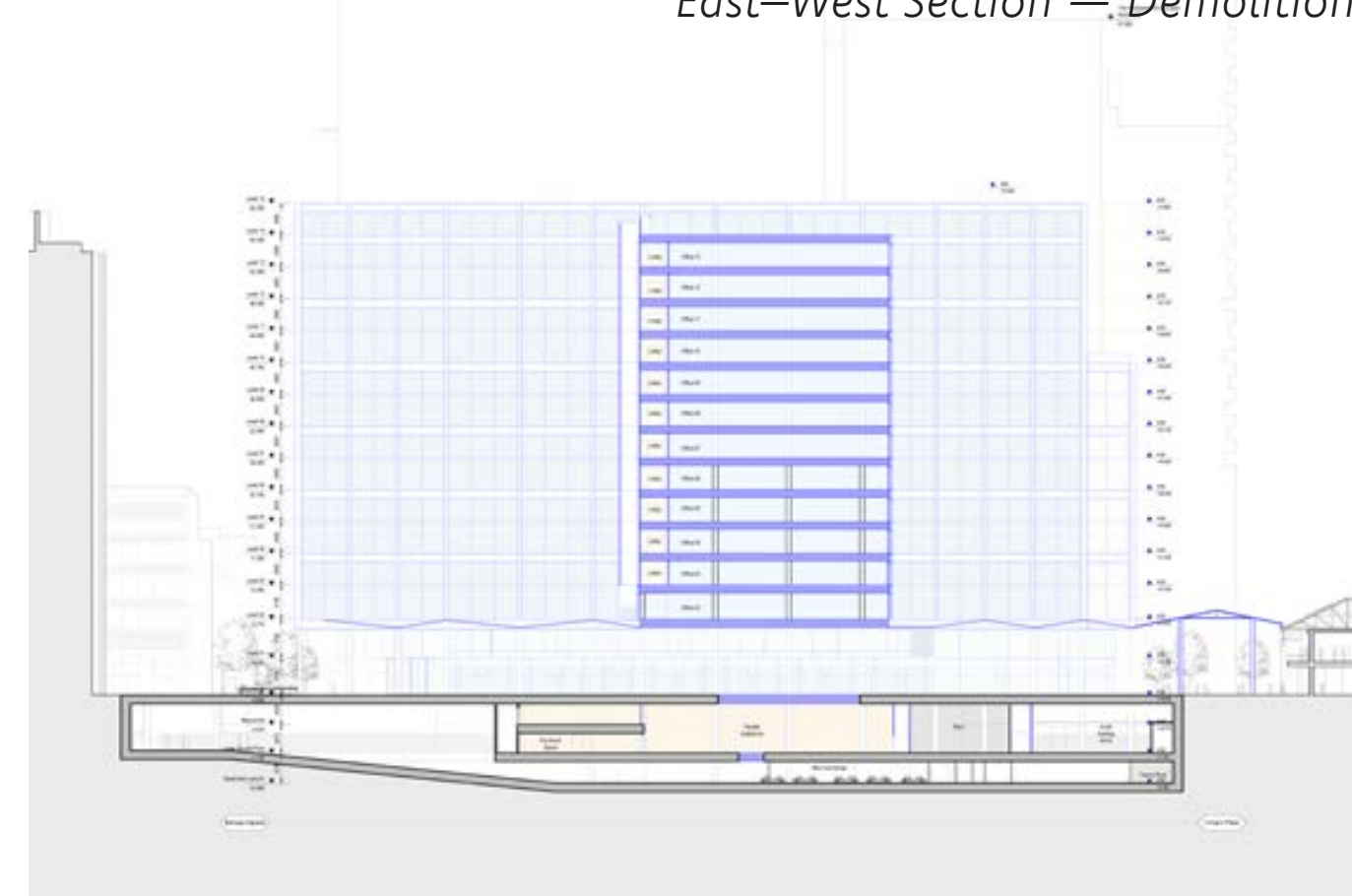
North–South Section — Existing



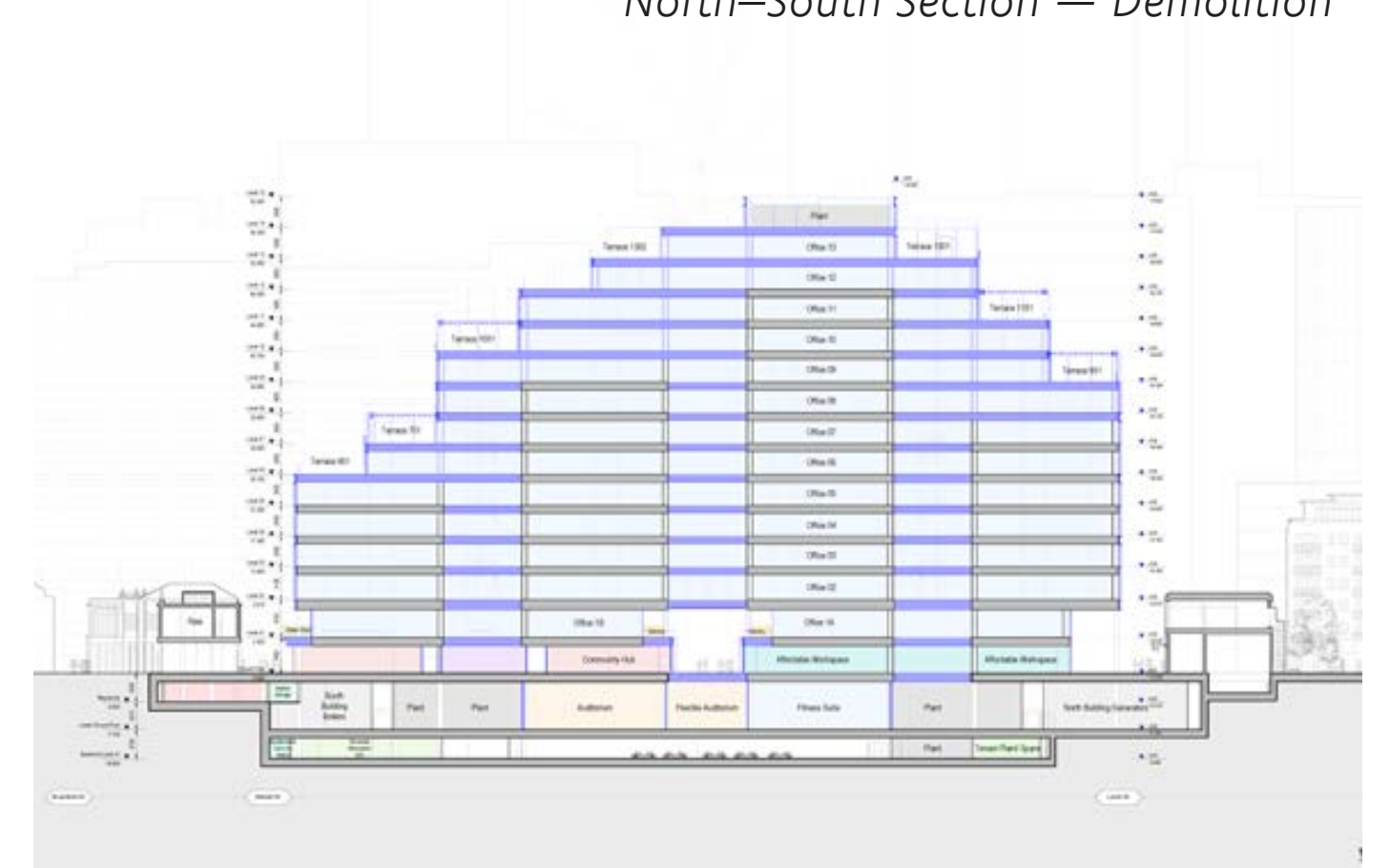
East–West Section — Demolition



North–South Section — Demolition



East–West Section — Proposed



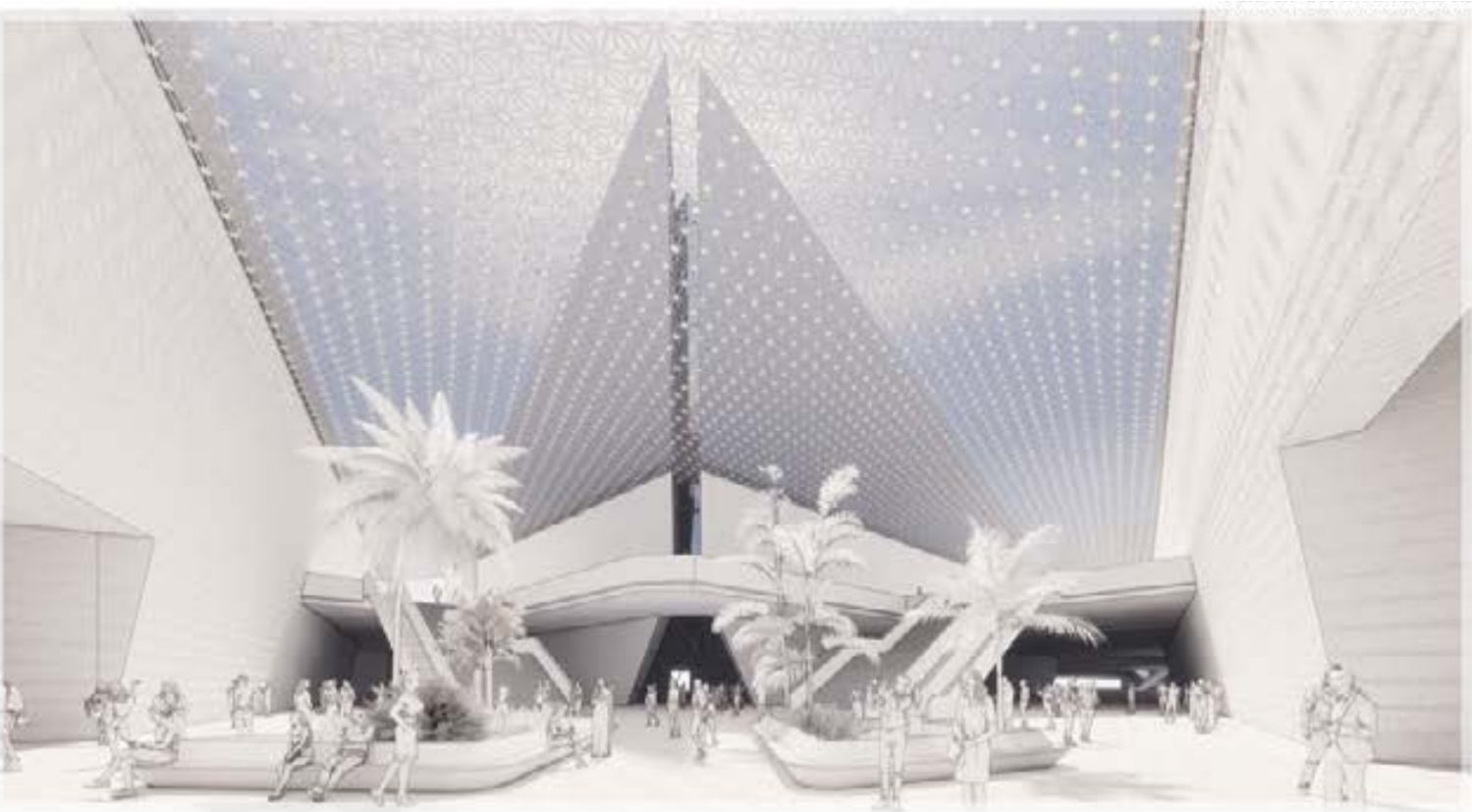
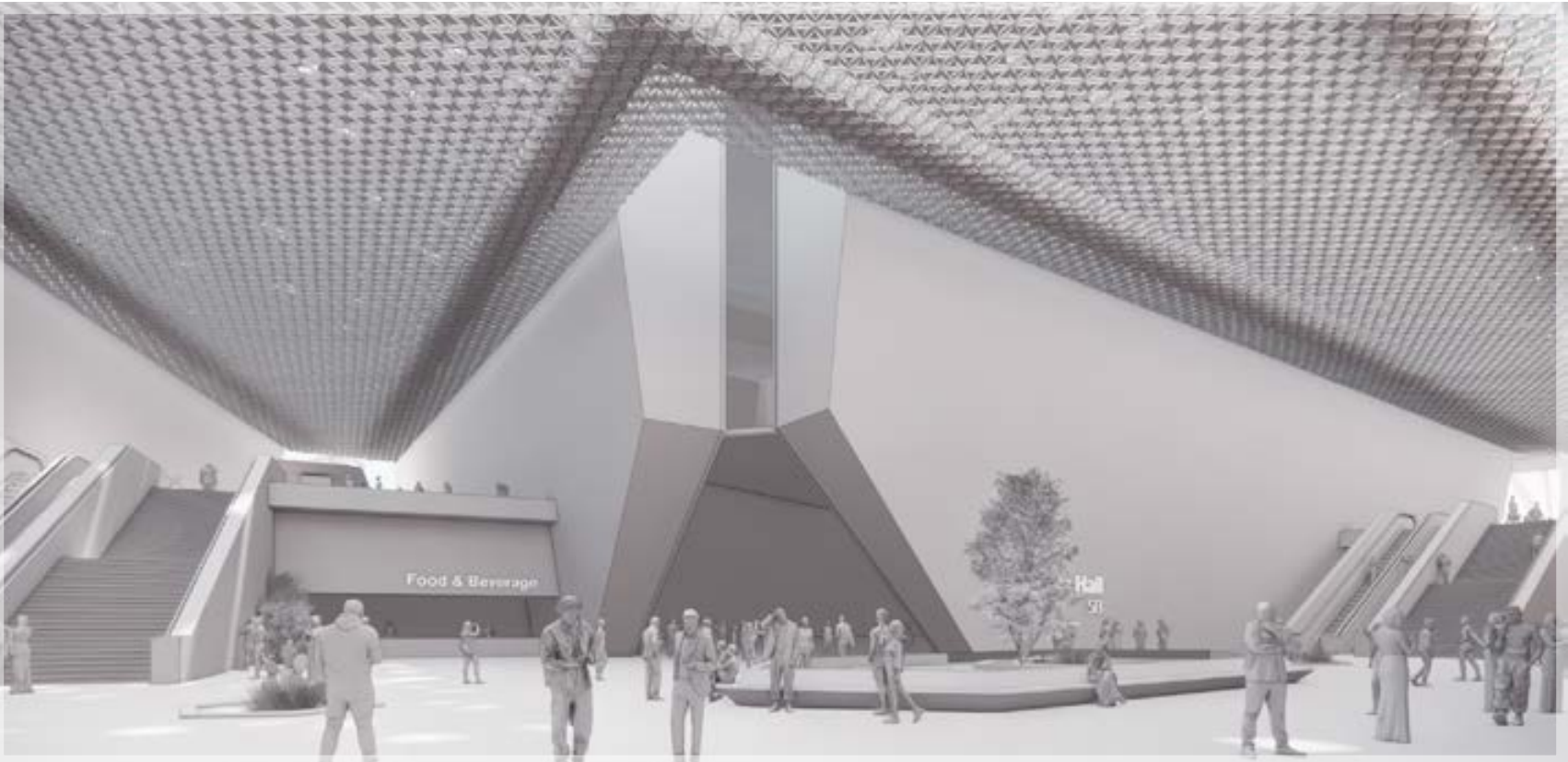
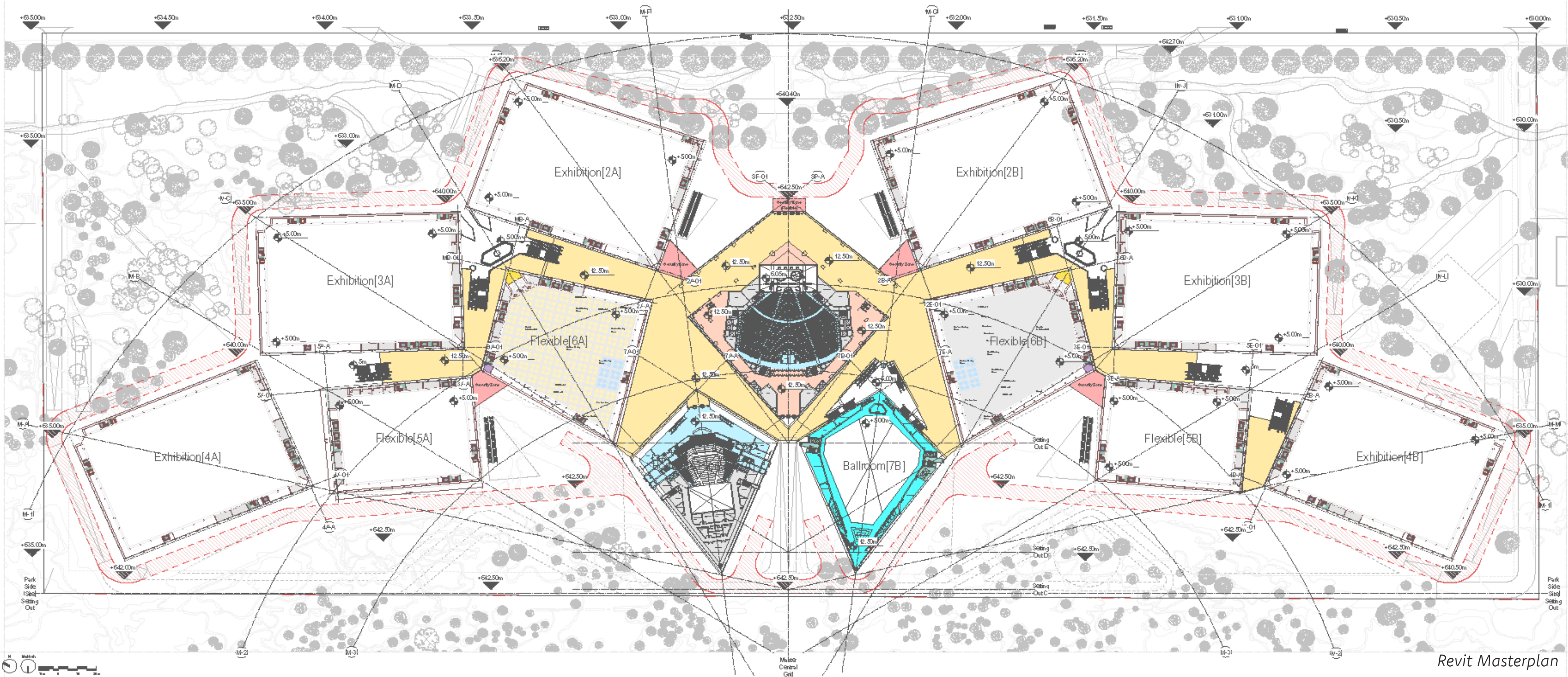
North–South Section — Proposed

RECC

REVIT, CAD, RHINO, GRASSHOPPER, ENSCAPE

At RIBA Stage 2, I worked within a team of more than twenty on the design development of this large exhibition complex. My contributions included building massing studies, adjustments to inclined walls and major spatial geometries, and the development of the mezzanine canopy and sunken communal areas.

I also coordinated design updates between workstreams and incorporated evolving decisions into Revit drawings. Alongside this work, I helped develop a Grasshopper-based shading system, testing geometric variations to inform the façade strategy.

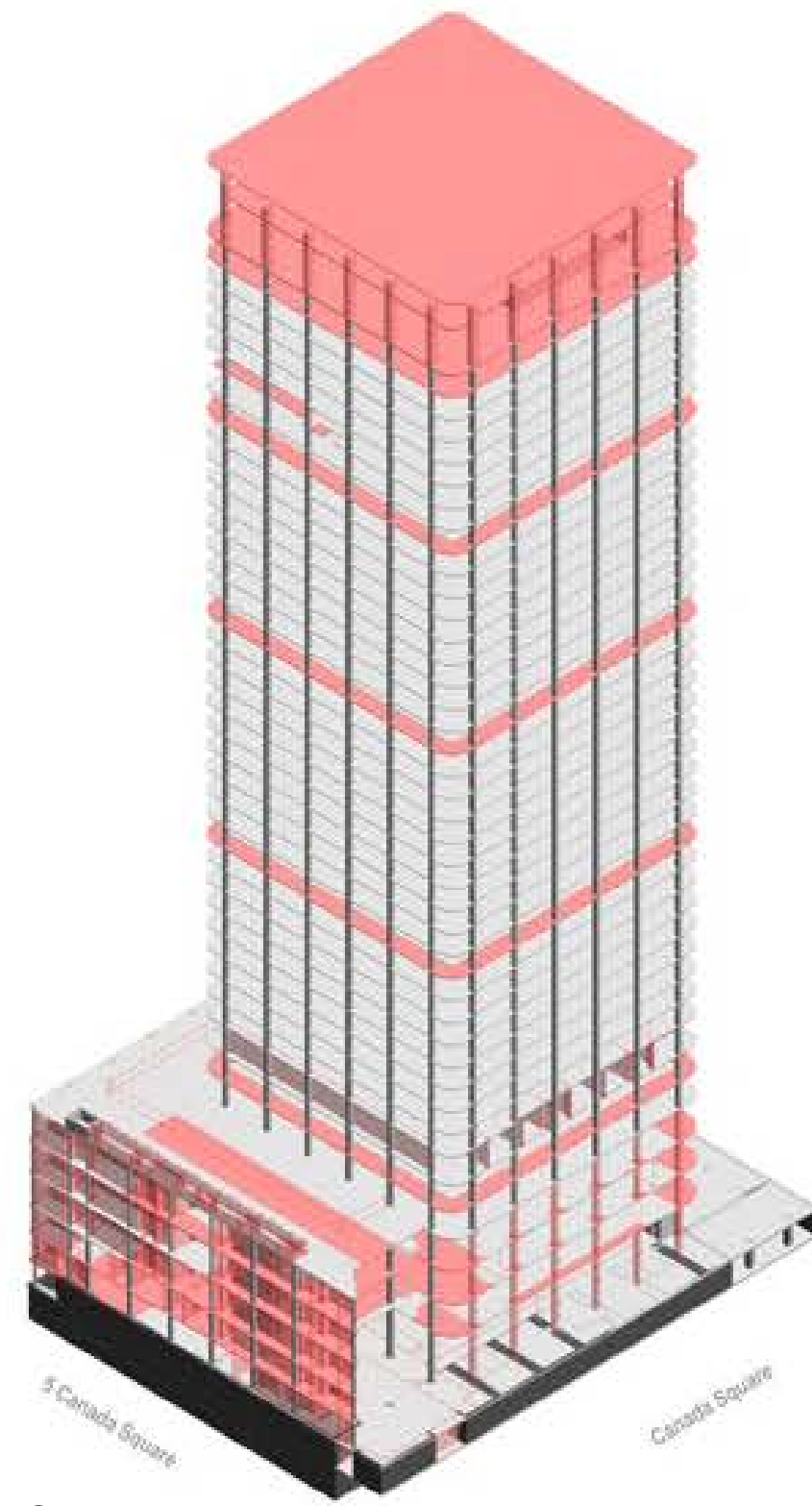


8 CANADA SQUARE

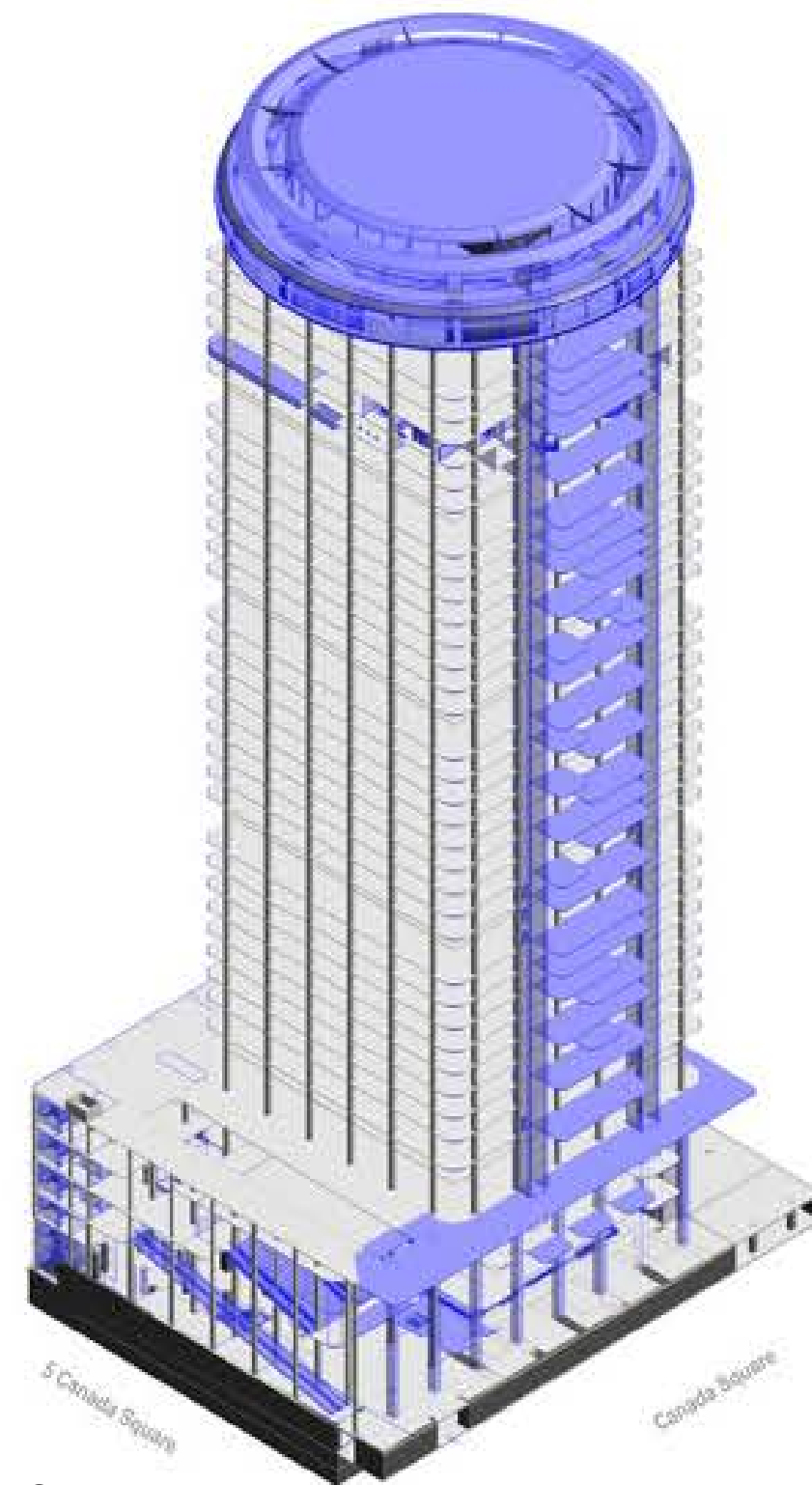
REVIT, RHINO, UNREAL ENGINE

This competition proposal explored the transformation of 8 Canada Square through selective retention and new intervention. Within the project team, I was responsible for Revit production and cinematic visualisation for the proposal, translating the evolving design into building models and real-time spatial sequences.

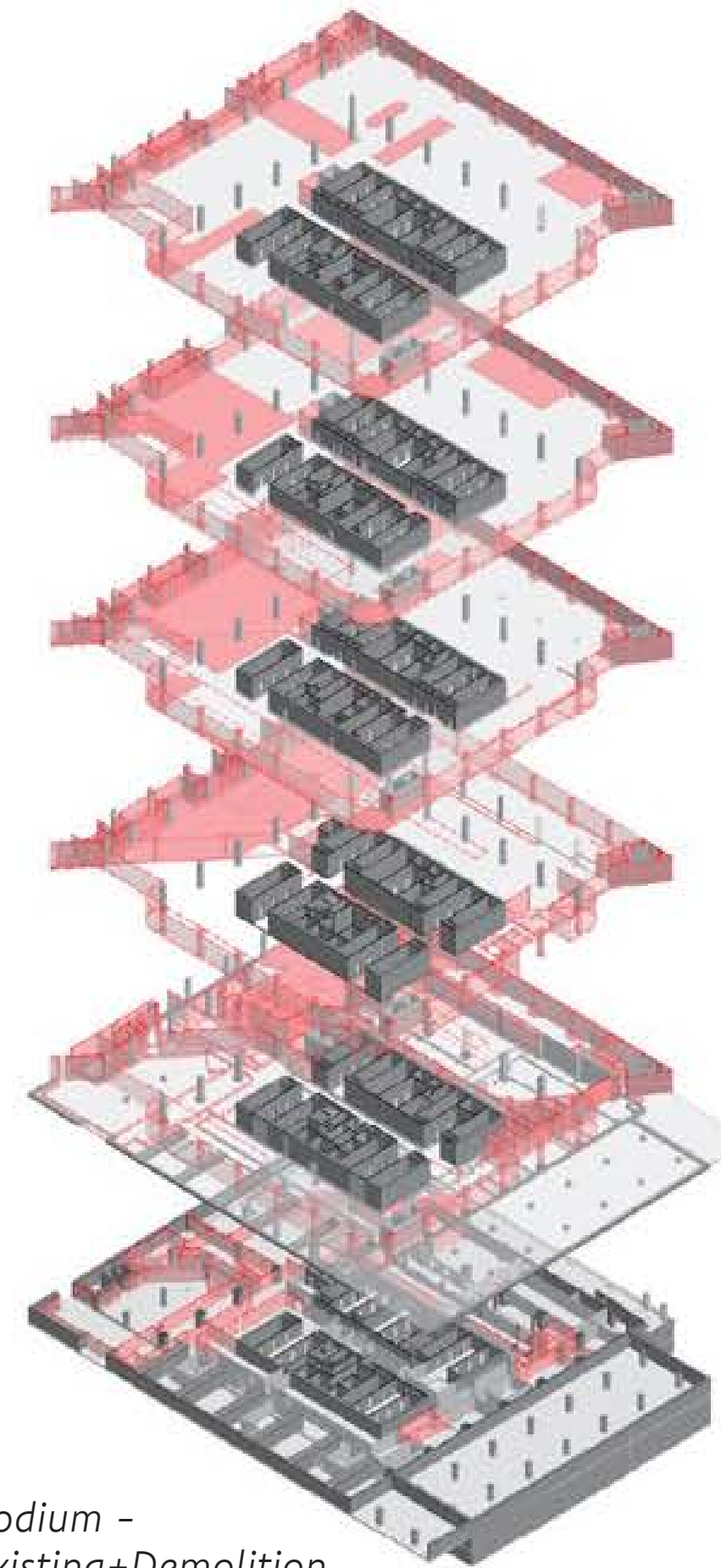
The proposed Tower Top introduced exhibition and observation spaces at the summit of the commercial tower. The visualisations explored how these spaces could create a public cultural destination, connecting interior experience with views across the city.



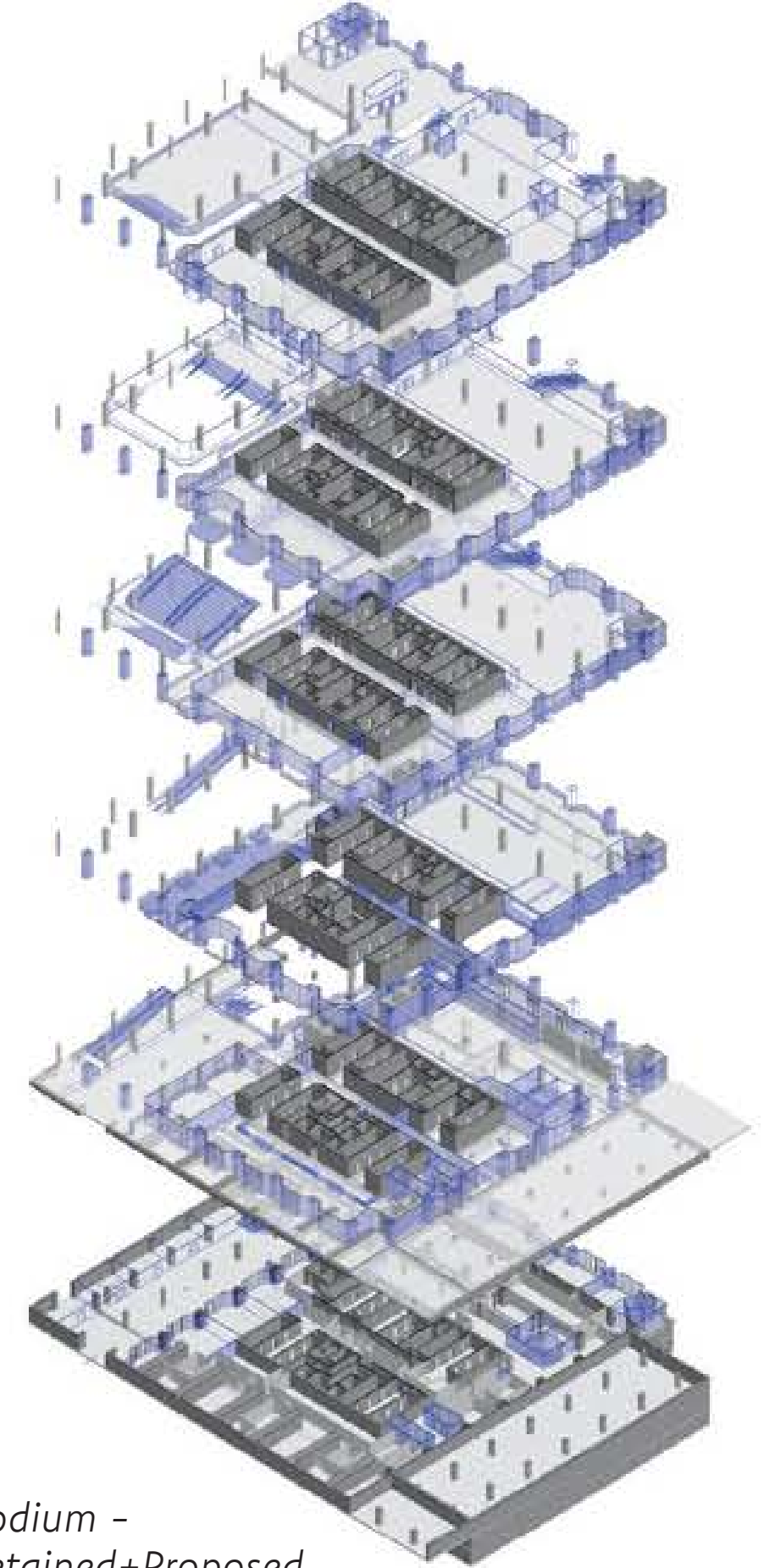
SW -
Existing+Demolition



SW -
Retained+Proposed



Podium -
Existing+Demolition



Podium -
Retained+Proposed

Podium - Exhibition Room



Tower Top - Observation Deck I



Tower Top - Observation Deck II



2. SACRED INTELLIGENCE

PROJECT OVERVIEW
SITE & CULTURAL RESEARCH
ROBOTIC MAKING
SPACES OF EXCHANGE
ARCHITECTURAL INTEGRATION

Sacred Intelligence explores how architecture can support the continuity and exchange of Tibetan Buddhist practices across distance. The proposal connects religious spaces at Gonggar with spaces in London through shared making, teaching, meditation and responsive imagery.

Drawing on spatial research, cultural dialogue and physical prototyping, the project develops architectural settings for these encounters. Robotic and media systems become part of the spaces through which practices are shared, exploring how remote participation can acquire a physical and collective presence.

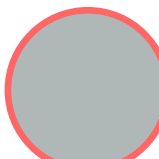
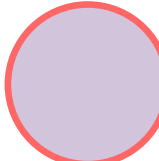
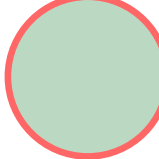
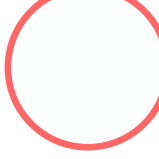


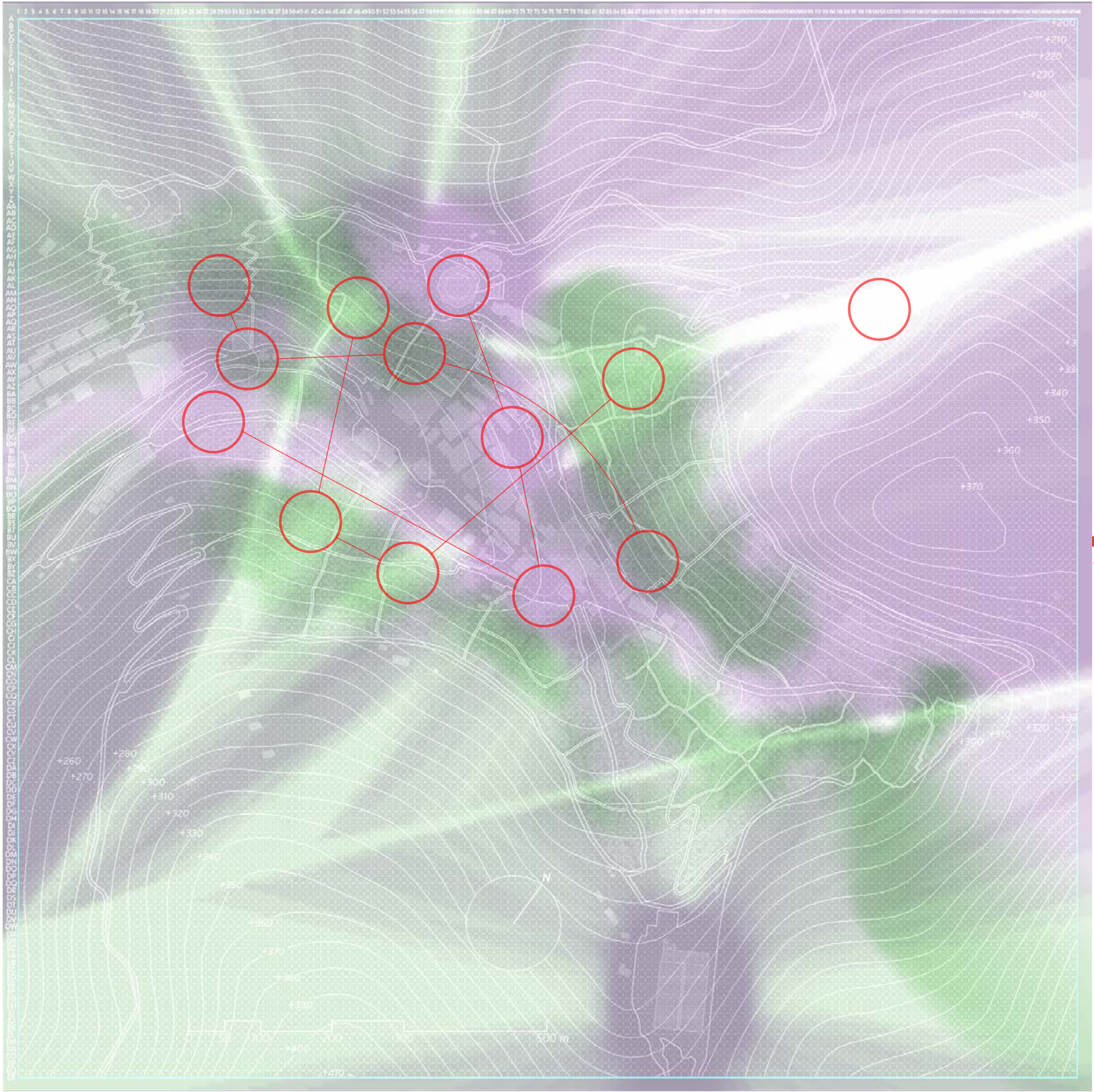
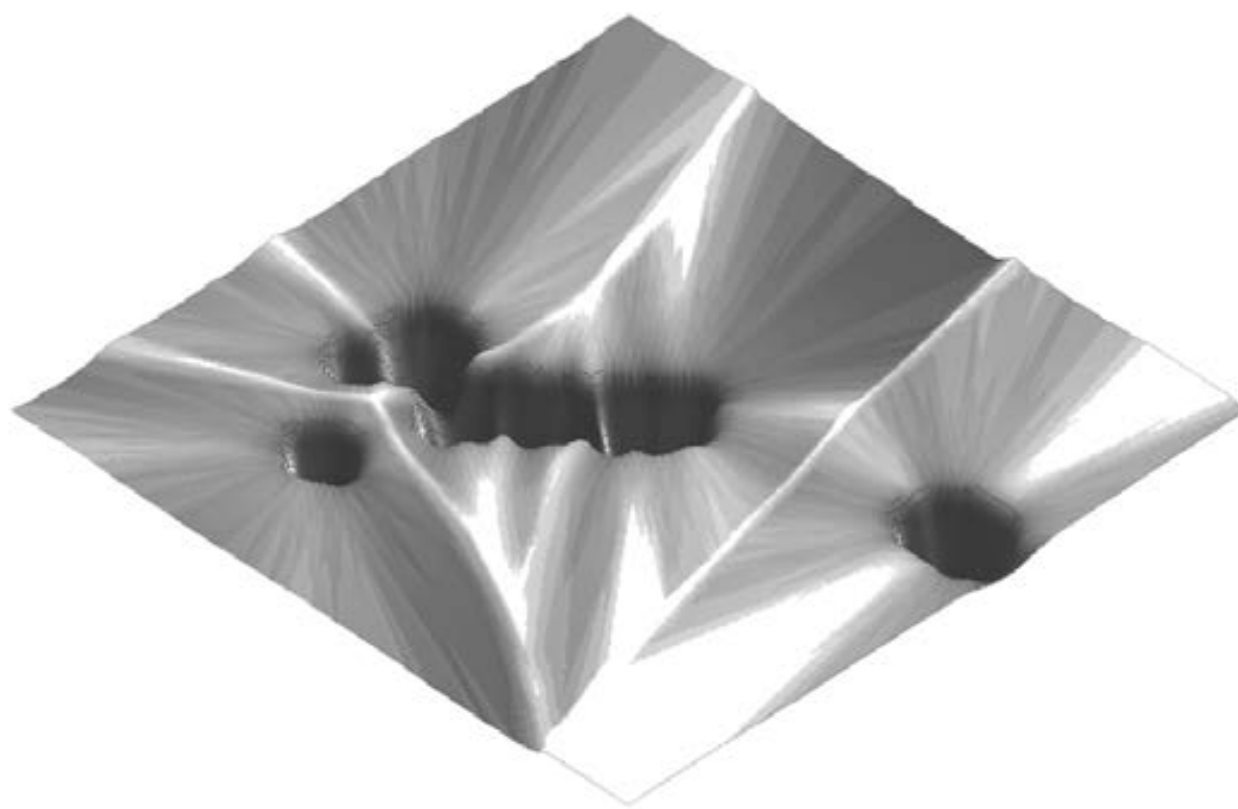
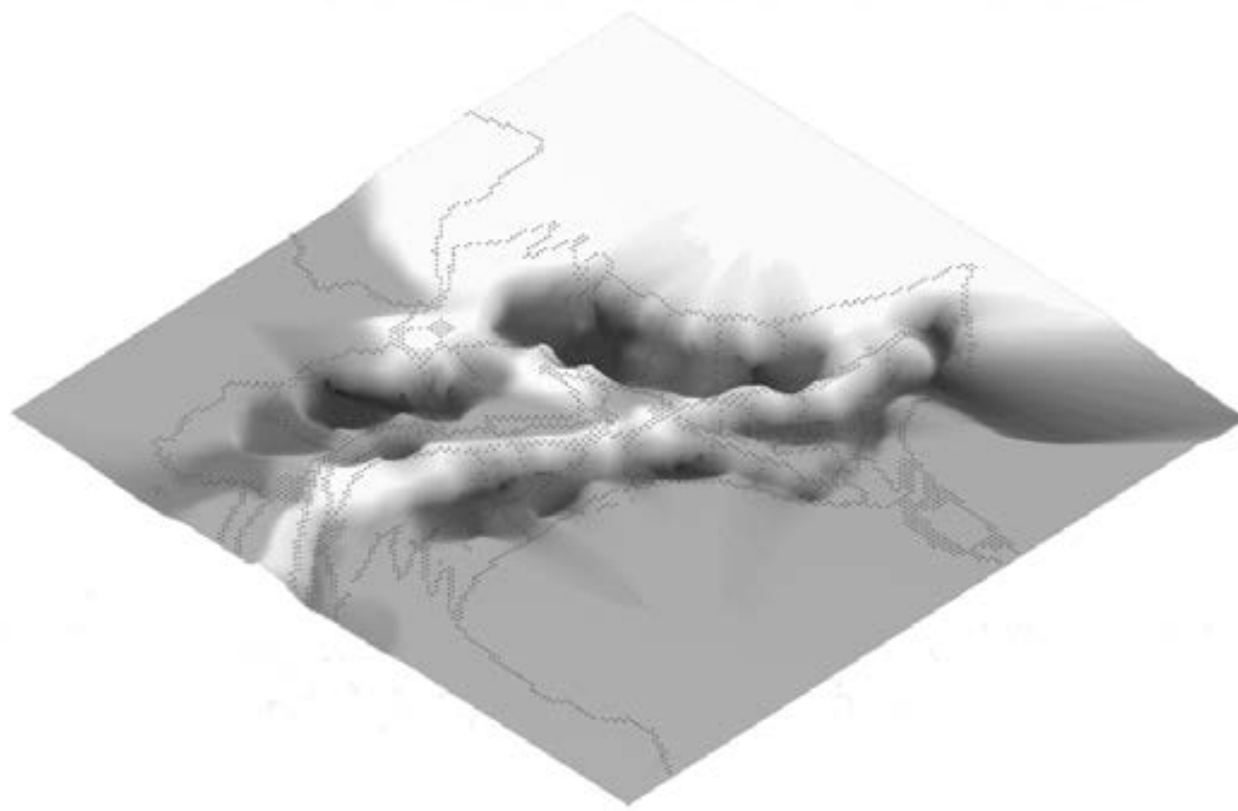
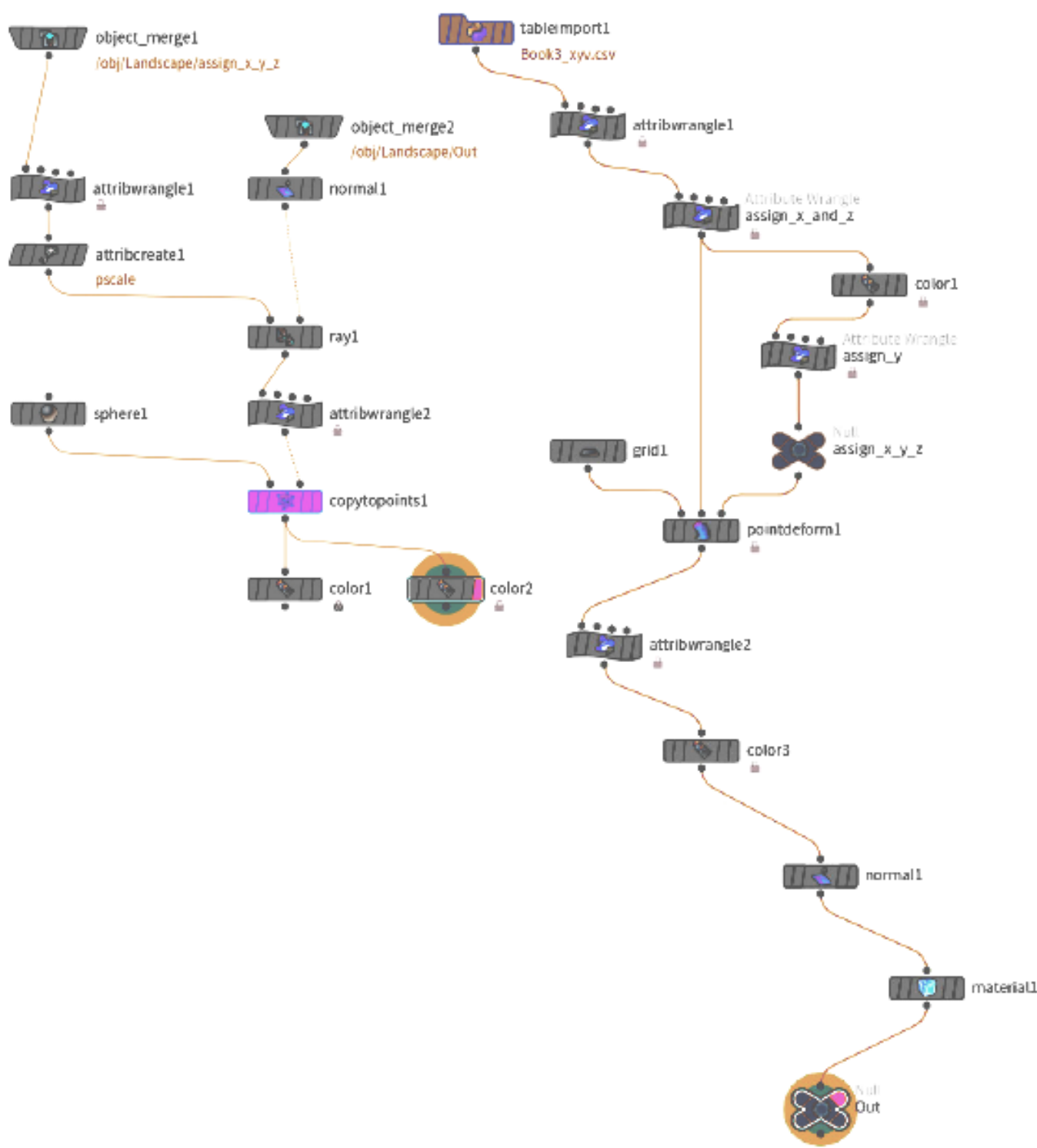
*A Mediated Assembly Hall
Technological Speculation on Tibetan Buddhist Learning Space*

The spatial study began with a reconstruction of Larung Gar from limited visual sources. More than 10,000 self-built dwellings were traced manually alongside roads, public buildings and religious institutions, establishing a base for spatial analysis.

The simulation explores where religious attraction and administrative constraints overlap. Weighted coordinate datasets represent the influence of major monastic institutions alongside boundaries, infrastructure and regulatory pressures. Their interaction produces fields of convergence and divergence, allowing different spatial priorities to be compared across the site.

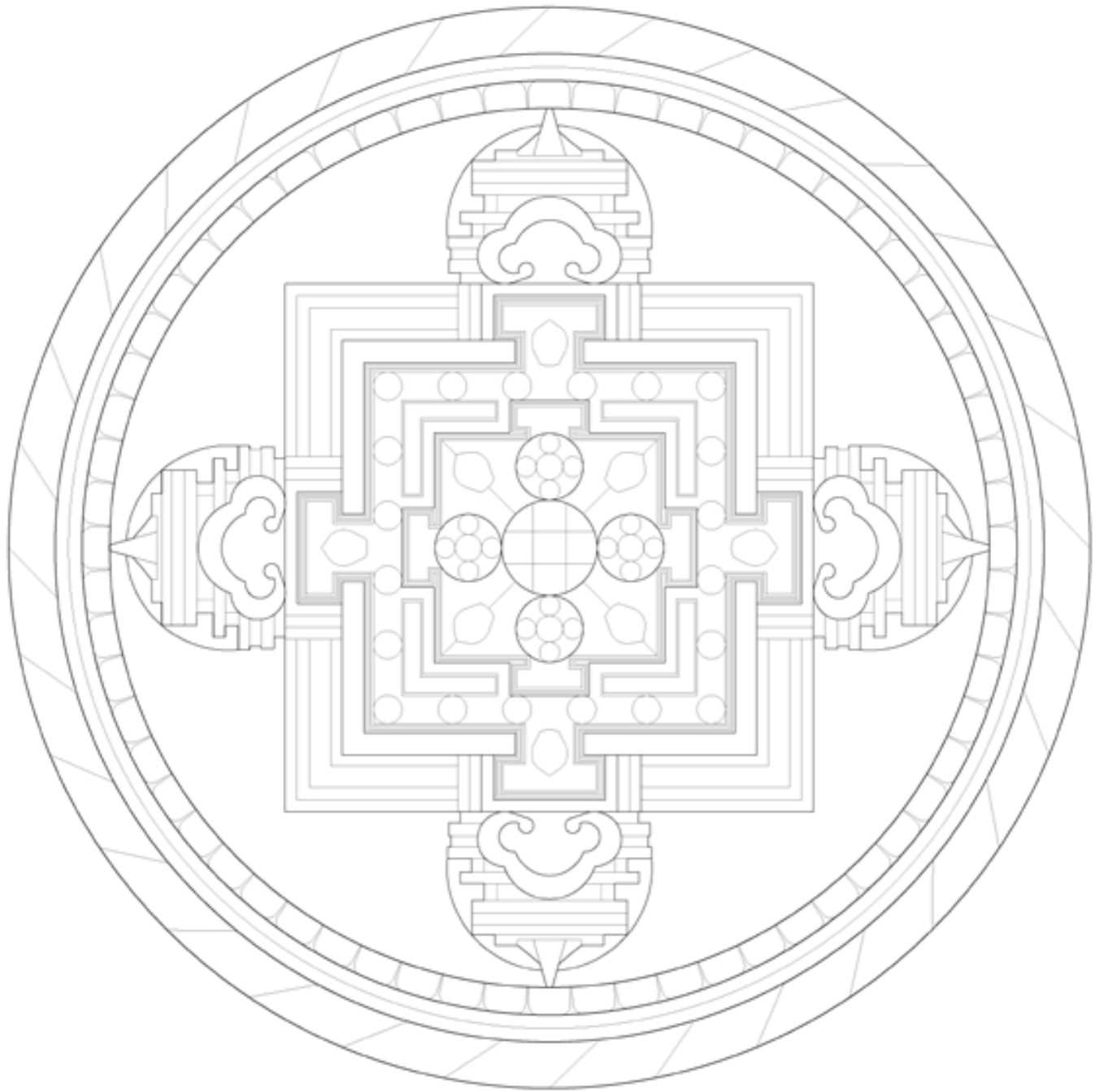
Read as exploratory scenarios shaped by the model’s assumptions, these outputs identify candidate areas where competing priorities may create opportunities or constraints for architectural intervention.

-  REGIONS OF INTEREST TO **BOTH SIDES**
-  AREAS OF INTEREST ONLY TO **BUDDHISTS**
-  REGIONS PREFERRED ONLY BY **ADMINISTRATIVE FORCES**
-  REGIONS OF INTEREST TO **NEITHER SIDE**



CULTURAL RESEARCH

LEARNING FROM THE FIVE SCIENCES

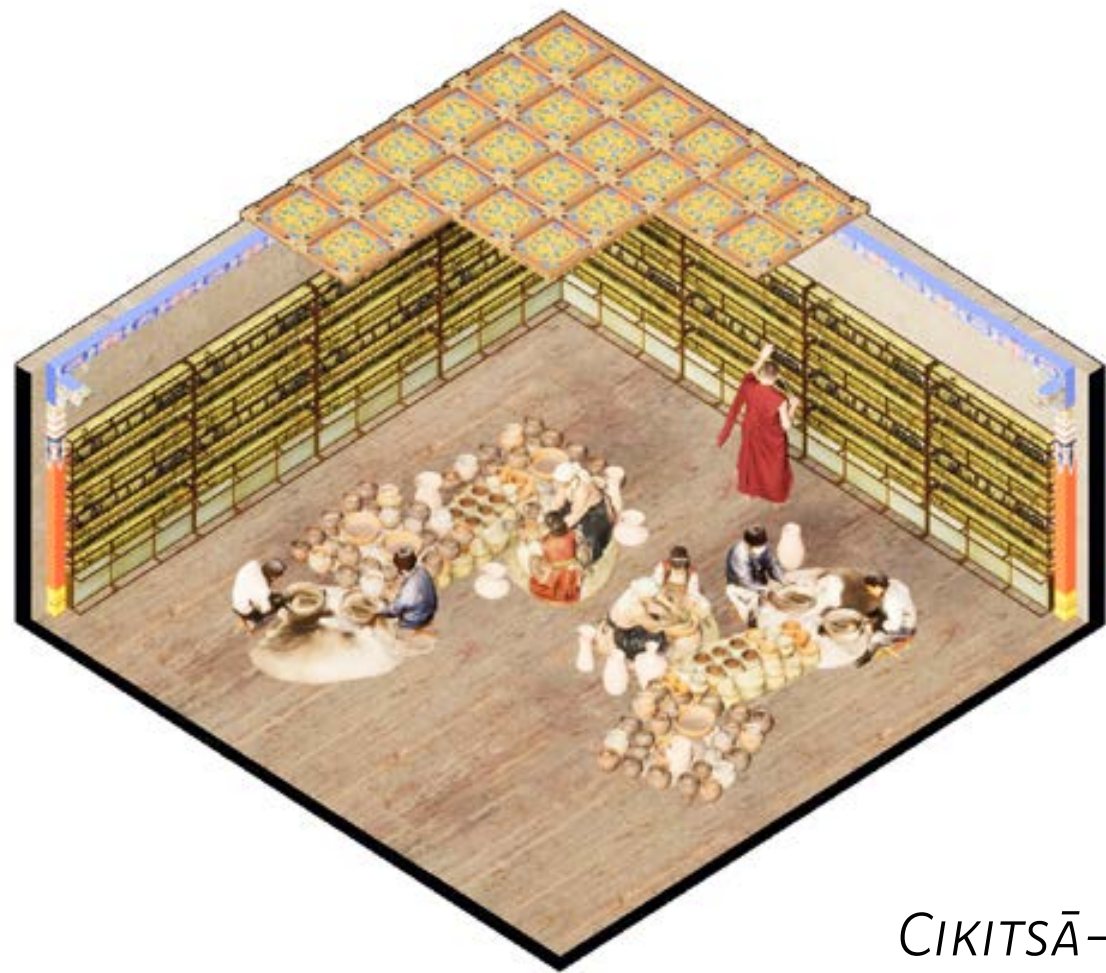


The “Five Sciences” provides a framework for studying Tibetan Buddhist knowledge and practice. Within this wider field, the project focuses on the sand mandala, examining the relationship between its geometric organisation, ritual significance and process of making.

Its constituent forms are analysed and classified in preparation for algorithmic transcription. This study asks what aspects of the practice can be translated into a computational process, and how that translation might support its continuation and exchange.



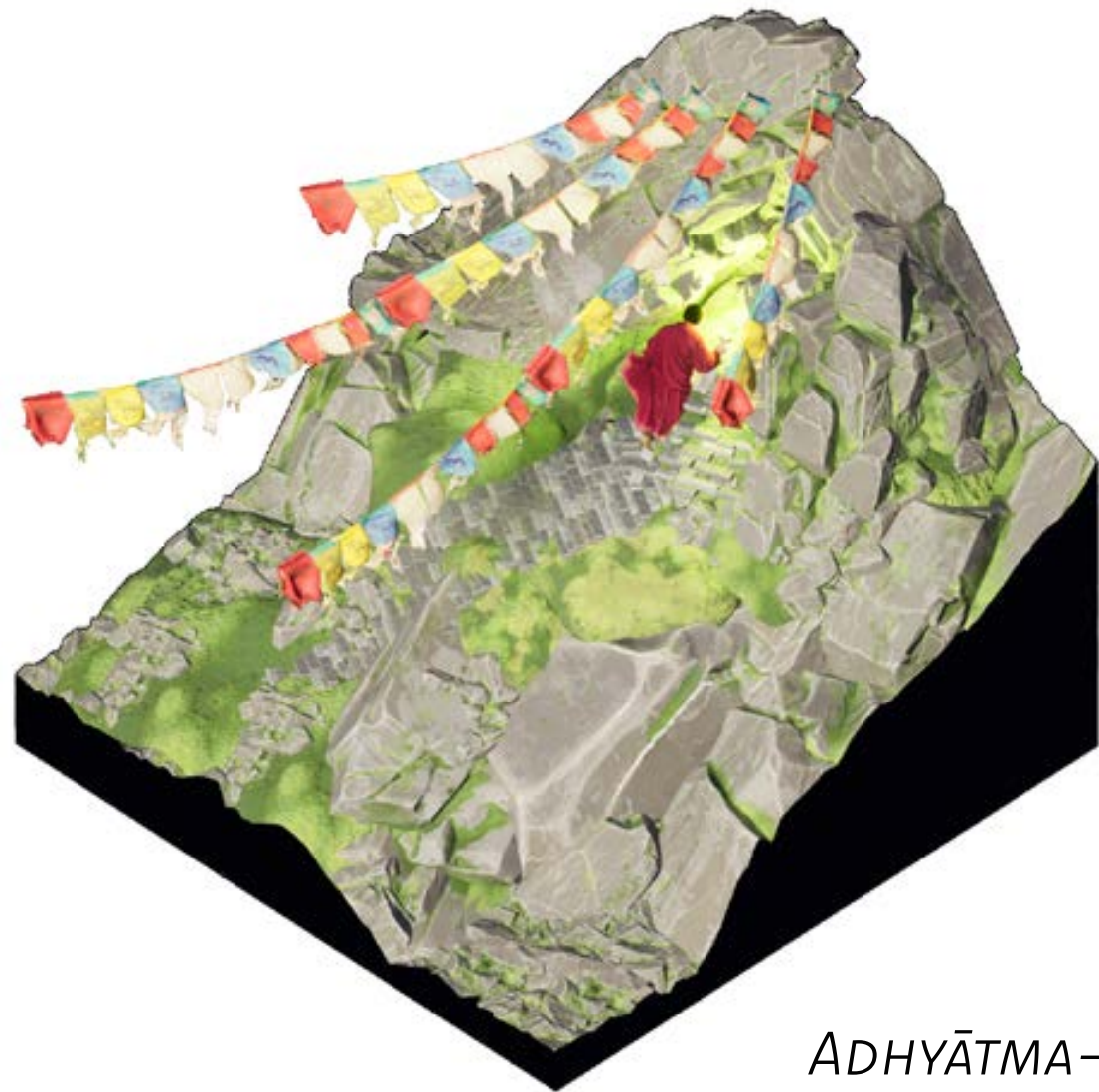
ŚILPA-VIDYĀ
"SCIENCE OF ARTS AND CRAFTS"



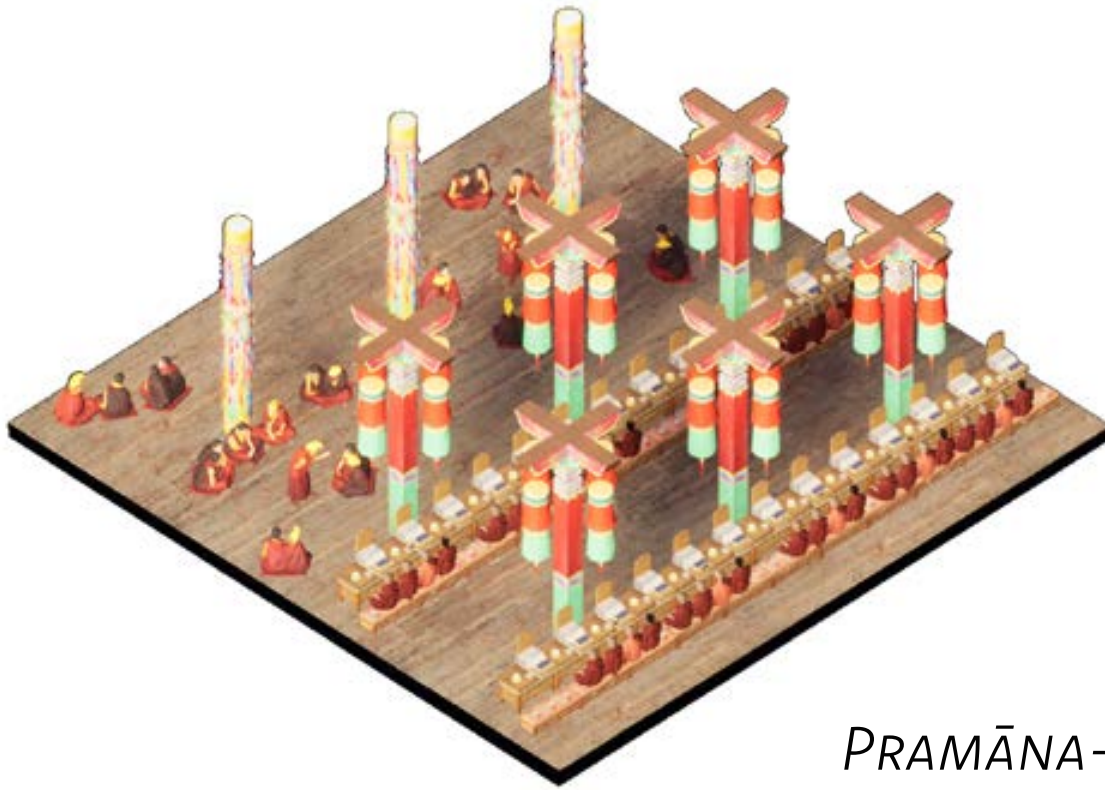
CIKITSĀ-VIDYĀ
"SCIENCE OF MEDICINE"



ŚABDA-VIDYĀ
"SCIENCE OF LANGUAGE"



ADHYĀTMA-VIDYĀ
"INNER SCIENCE"



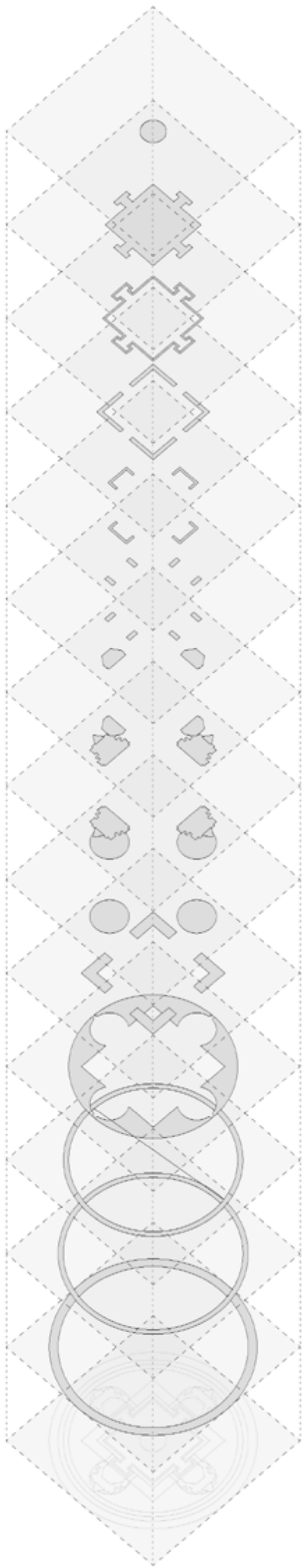
PRAMĀNA-VIDYĀ
"SCIENCE OF LOGIC"

Fieldwork in Tibet brought the project into dialogue with senior Buddhist teachers who expressed an interest in sharing religious knowledge and practice across geographical boundaries. This prompted an investigation into how the making of a sand mandala might be transmitted between distant spaces.

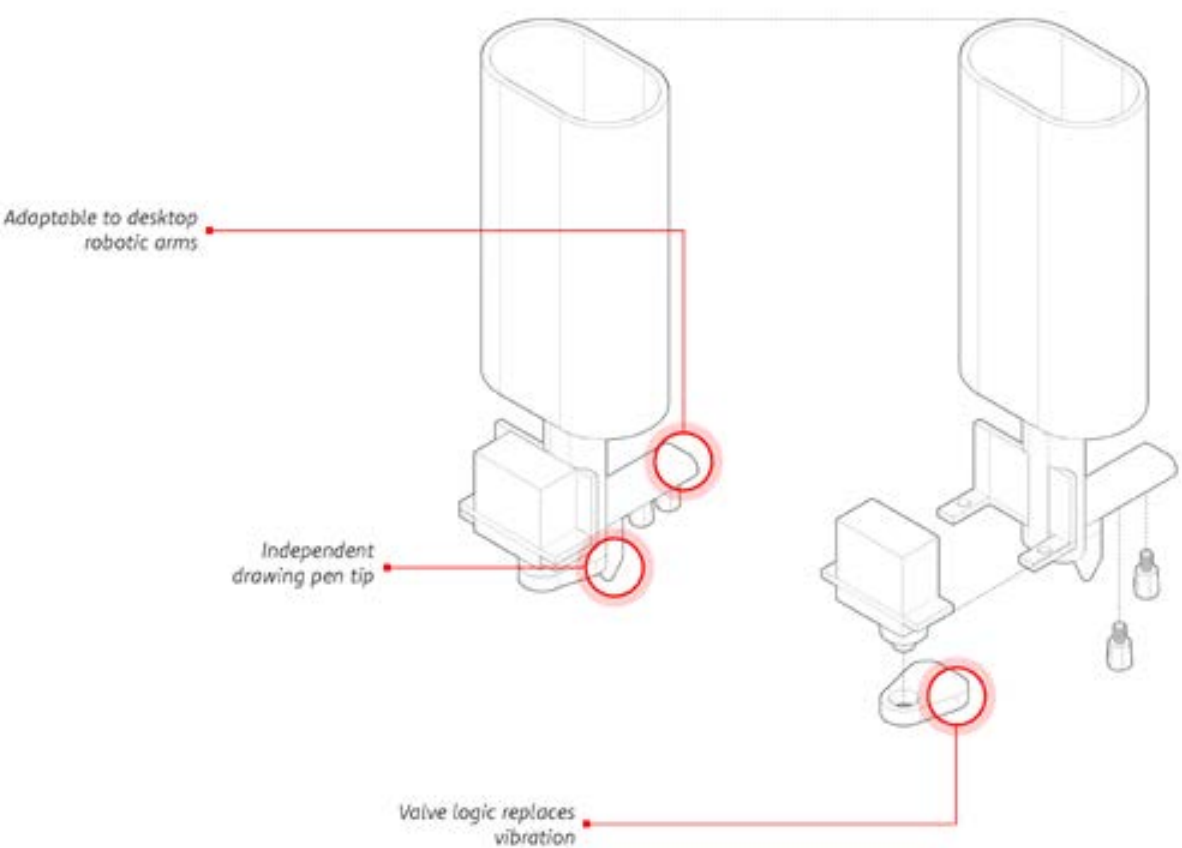
The diagrams decompose the mandala into geometric elements and classify them into layers, preparing its visual structure for algorithmic transcription. The hardware studies address the physical execution of the drawing, exploring sand delivery, colour separation and line formation.

Successive prototypes developed and tested different dispensing mechanisms. The resulting sand-deposition head provides a physical basis for the remote ritual proposal developed in the following pages.

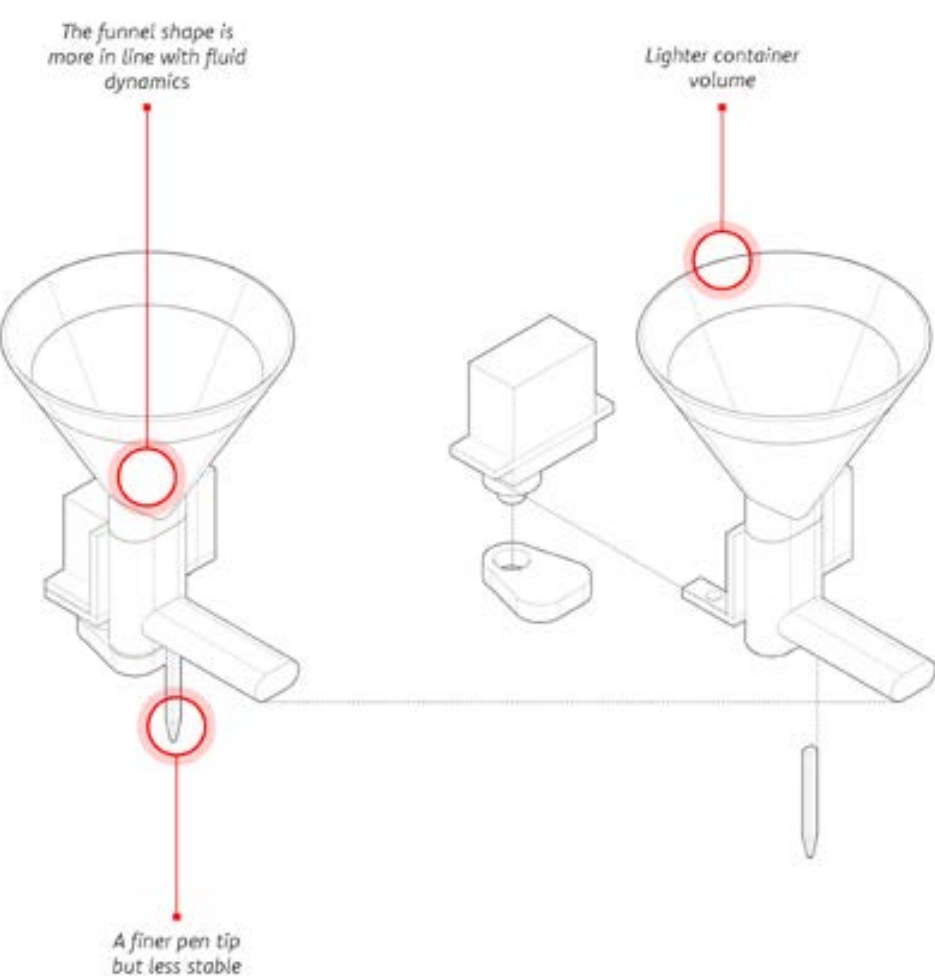
LAYER CLASSIFICATION FOR ALGORITHMIC TRANSCRIPTION



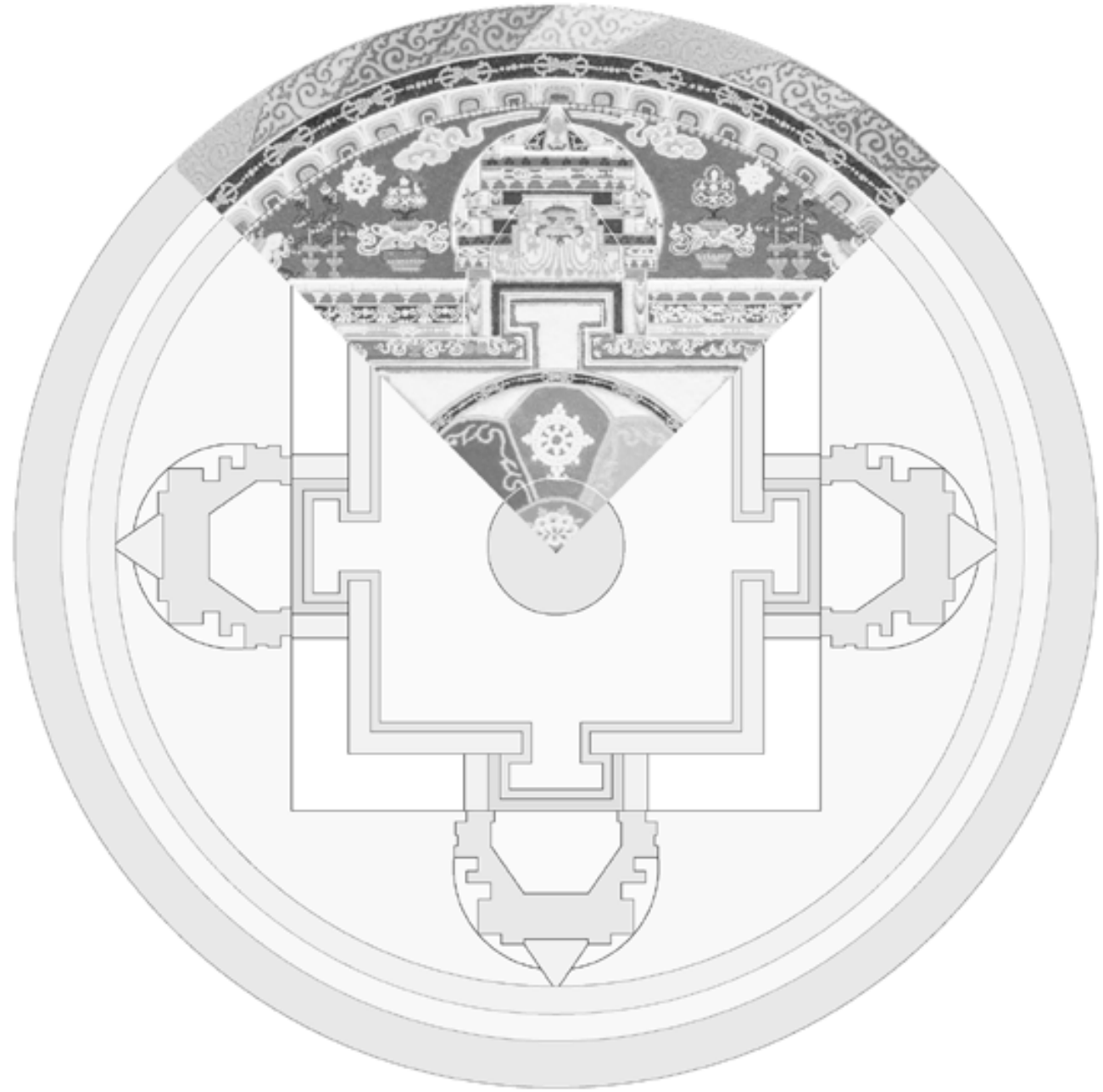
PROTOTYPE 6 - VALVE-CONTROLLED SAND DISPENSER



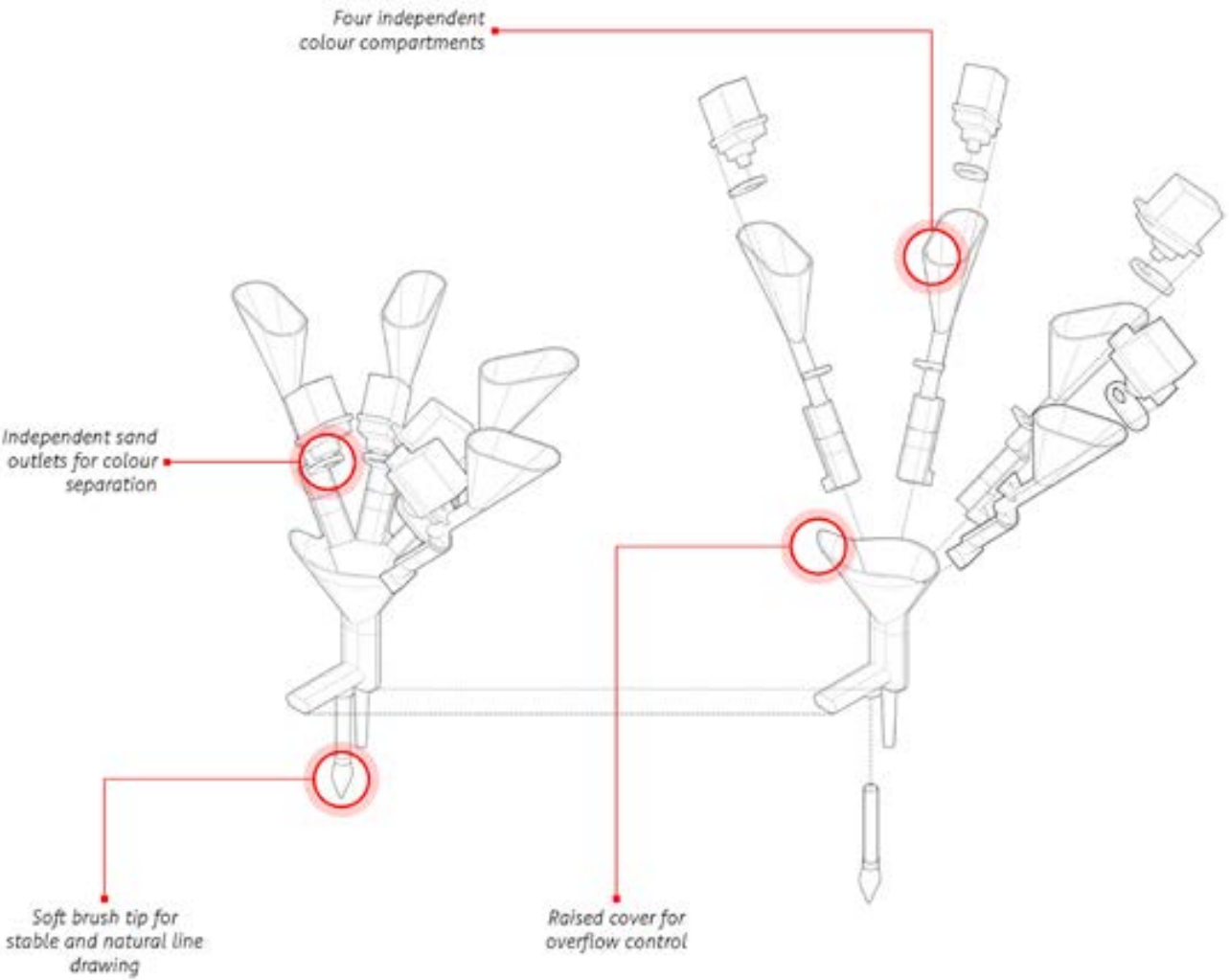
PROTOTYPE 7 - FUNNEL-BASED SAND DISPENSER



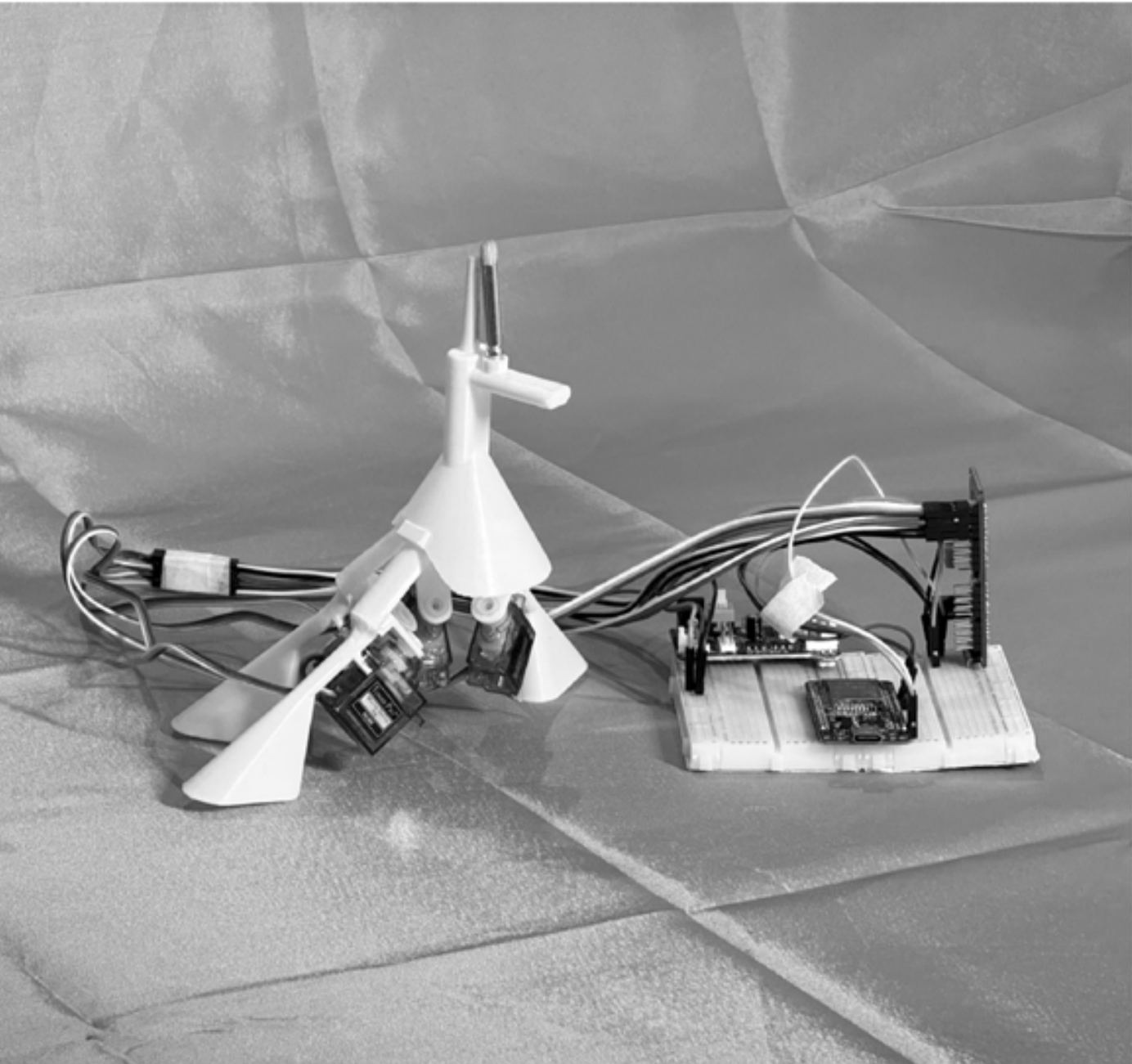
MANDALA GEOMETRY DECOMPOSITION



PROTOTYPE 8 - ROBOTIC SAND DEPOSITION HEAD



PROTOTYPE 8 - PHYSICAL MODEL



SPACES OF EXCHANGE

HALL OF MAKING
CLOISTER OF EXCHANGE
HALL OF RESONANCE
HALL OF REVELATION

The proposal develops four spatial responses to religious practice: the Hall of Making, Cloister of Exchange, Hall of Resonance and Hall of Revelation. Together, they explore remote making, teaching, collective meditation and the transmission of knowledge through imagery.

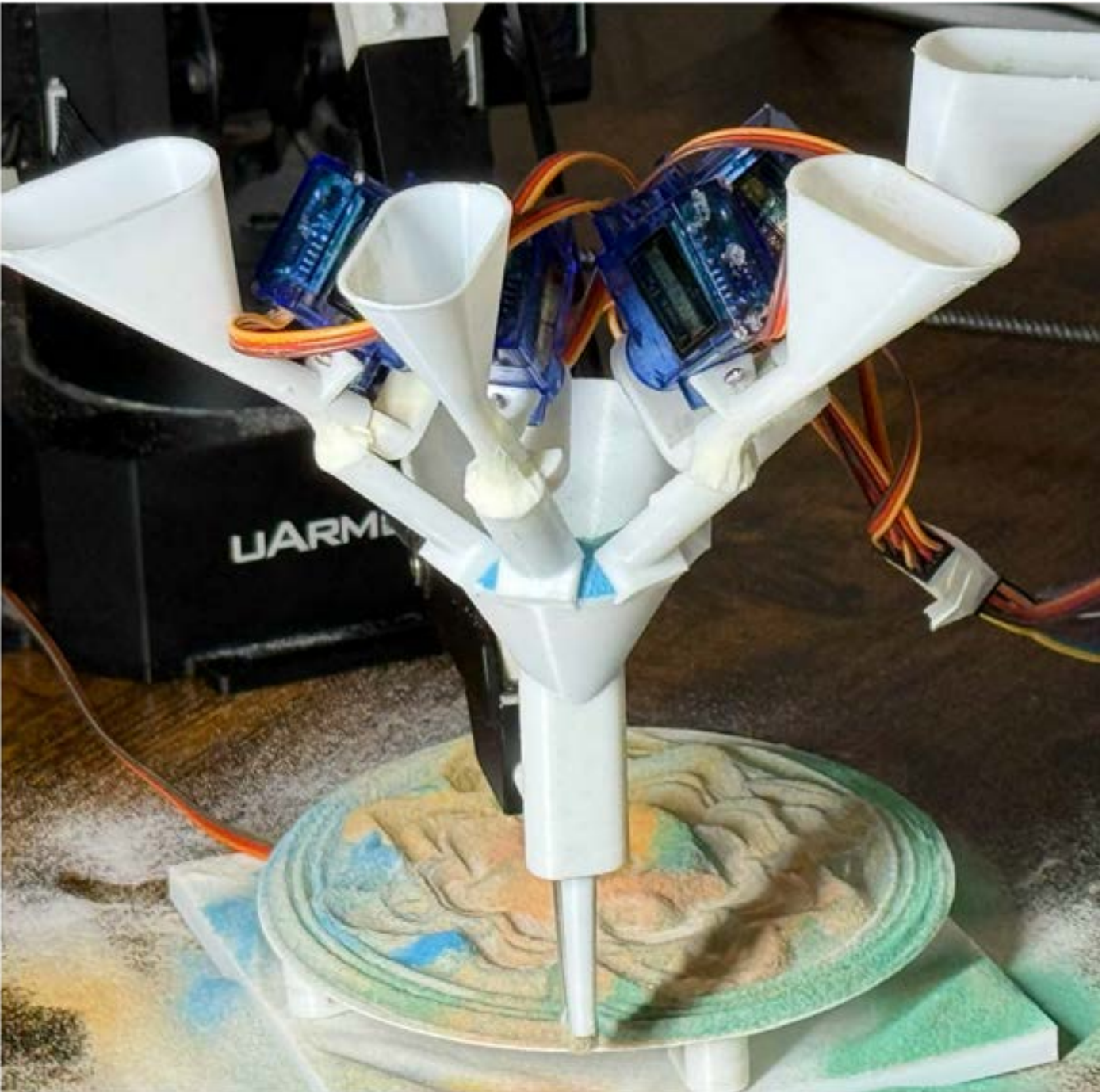
Two systems were developed as operational 1:1 prototypes and presented to senior Buddhist teachers during fieldwork in Tibet. These tests and discussions informed the proposed uses and their relationship to the spaces designed around them.



Robotic Sand Mandala

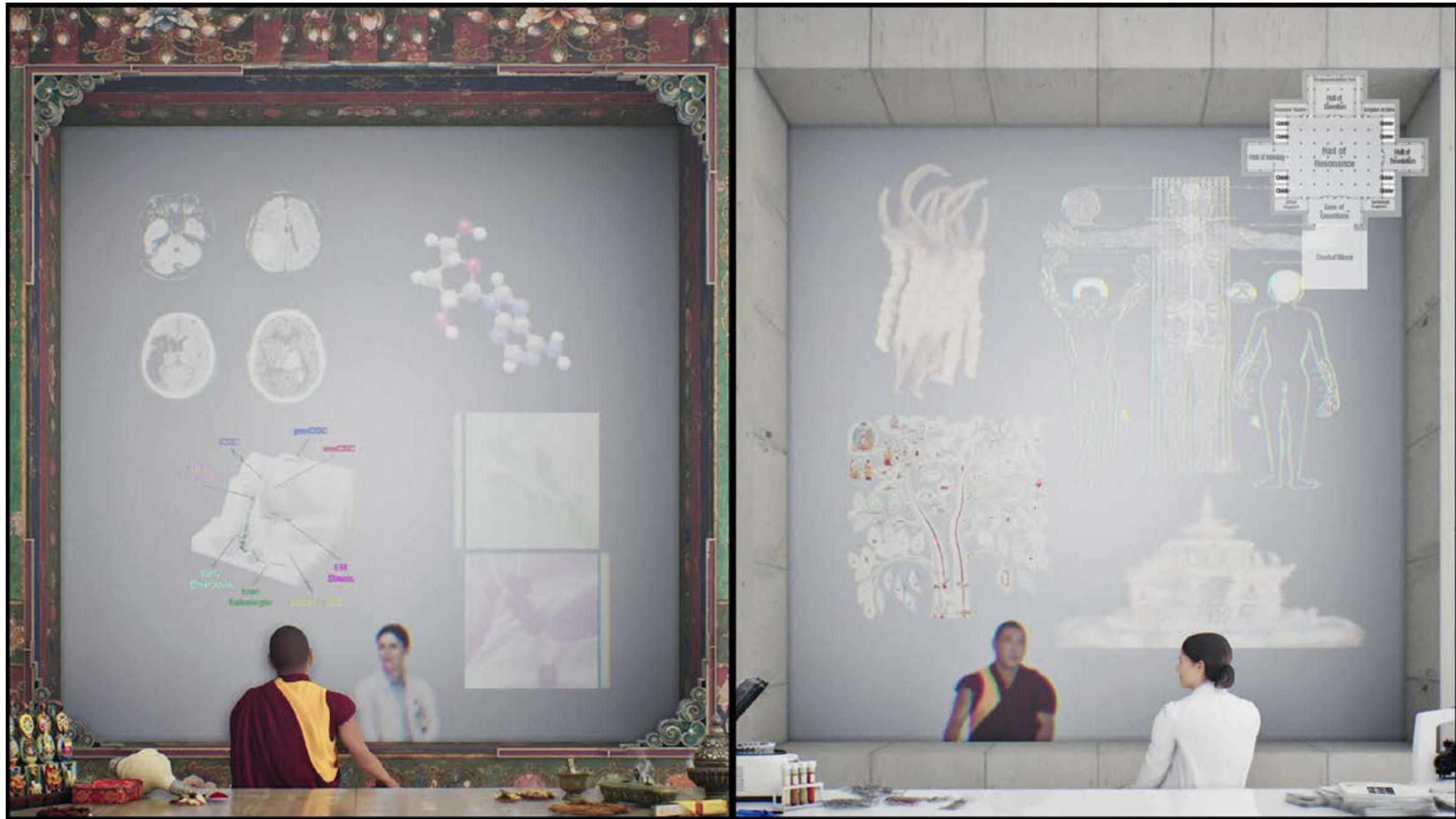
The Hall of Making proposes a connection through which a mandala master at Gonggar could transmit drawing instructions to a robotic installation in London. Participants at the receiving site would gather around the making process as the mandala takes shape, giving remote exchange a physical and spatial presence.

Scan / click
Prototype test video



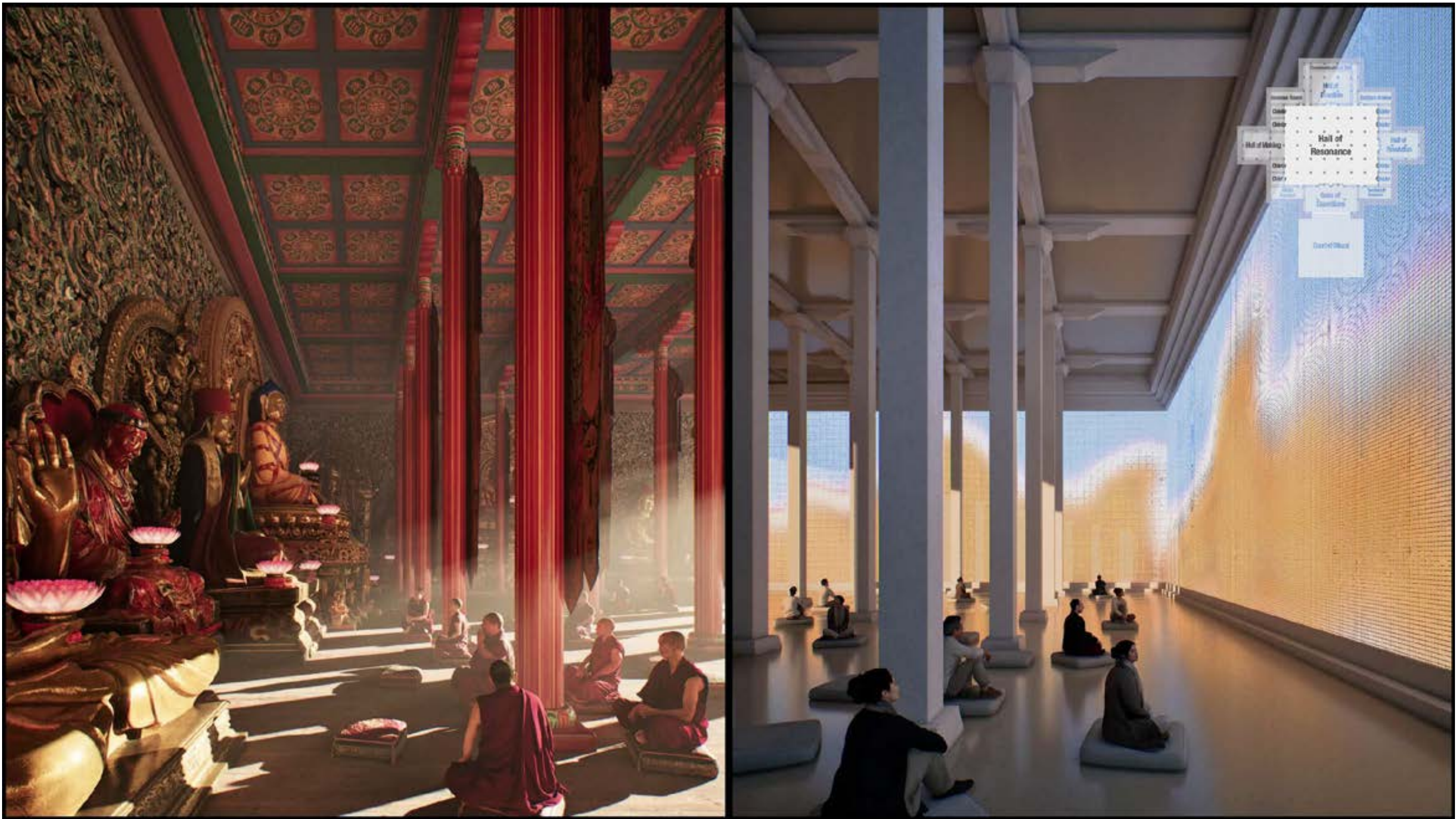
Multi-Colour Sand Mandala Fabrication System

The 1:1 prototype combines four independent sand channels within a single deposition head. Separate outlets control colour delivery, while a soft brush tip assists line formation. Physical tests explored how these components could work together during robotic drawing and informed successive adjustments to the mechanism.



Cloister of Exchange

The Cloister of Exchange provides a shared setting for teaching, discussion and study between distant monastic communities. Participants gather within corresponding spaces, where communication is integrated into the architecture and collective routines of learning. The proposal explores how a remote encounter can become part of a place's everyday use.



Hall of Resonance

The Hall of Resonance explores a connection between distant groups engaged in collective meditation. The proposed system translates breathing, presence and environmental inputs from one location into subtle spatial changes in the other. These changes are intended to make the presence of a remote community perceptible through a shared atmosphere.

AI-Driven Responsive Mural

The proposed system processes spoken teachings and contextual inputs from Gonggar to generate evolving mural imagery in London. The imagery is conceived as an interpretive layer accompanying the teaching, with the wall providing a spatial setting for its reception and discussion.



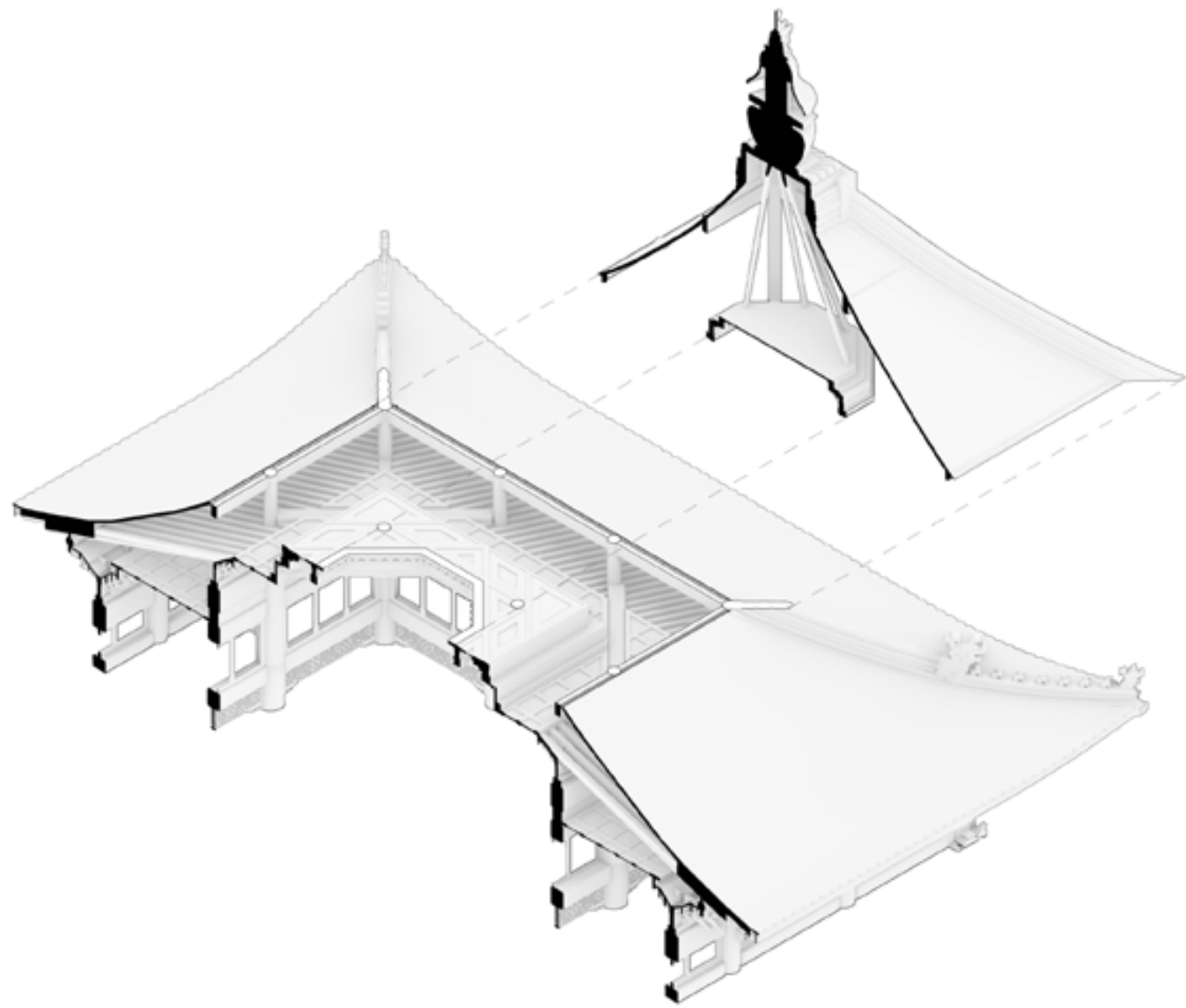
Hall of Revelation

The Hall of Revelation explores the architectural surface as a medium for the transmission of religious knowledge. It brings together spoken teaching, generated imagery and a responsive physical installation within a shared interior.



1:1 Architectural Media Prototype

The 1:1 prototype tests the relationship between dynamically generated imagery and a physical media surface. It explores how changing visual content can be incorporated into an architectural element and contribute to the experience of the surrounding space.



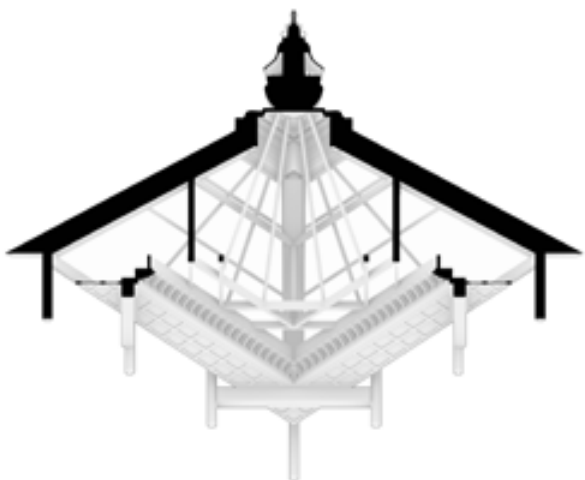
A THE ROOF STRUCTURE IS DIVIDED INTO TWO PARTS ACCORDING TO THE KEY FUNCTIONAL SECTIONS.

Architectural Adaptation

Studies of Tibetan Buddhist monastery spaces establish the proposal's spatial organisation, hierarchy and relationship to ritual use. The architectural development then examines how the remote mandala system could be accommodated within this framework.



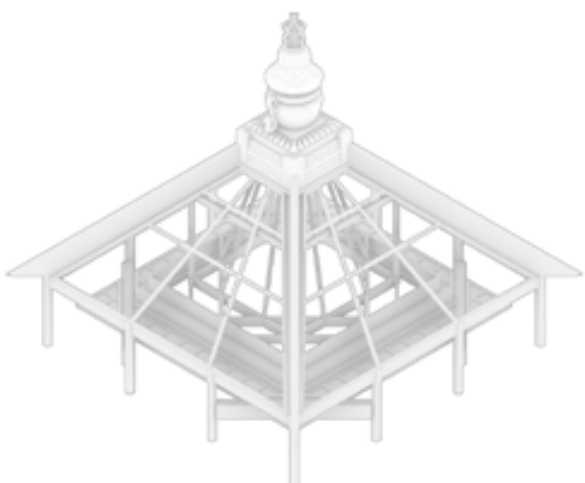
B SELECTED PART FOR RENOVATION



B' IMPROVED STRUCTURE

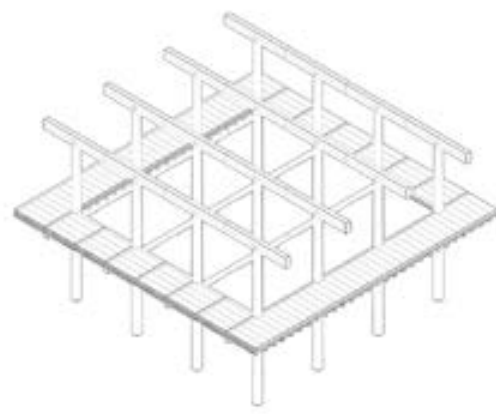


C SELECTED PART FOR RENOVATION

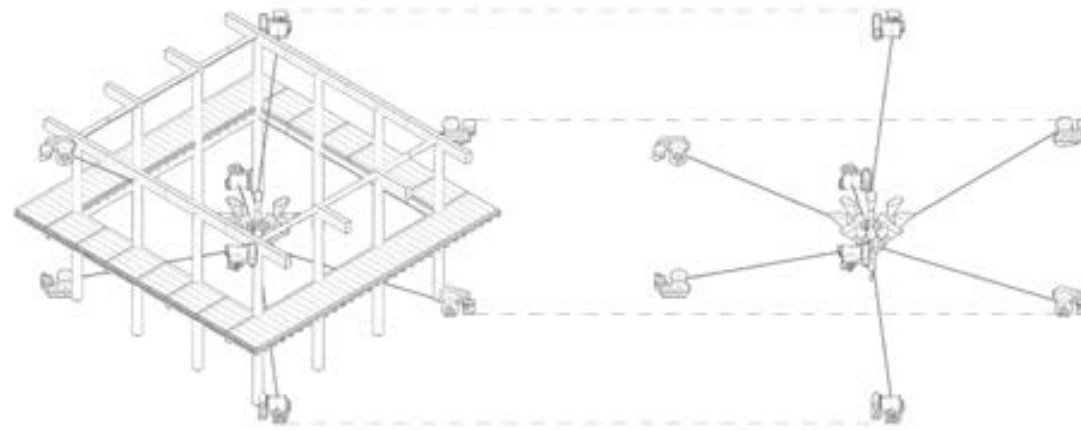


C' IMPROVED STRUCTURE

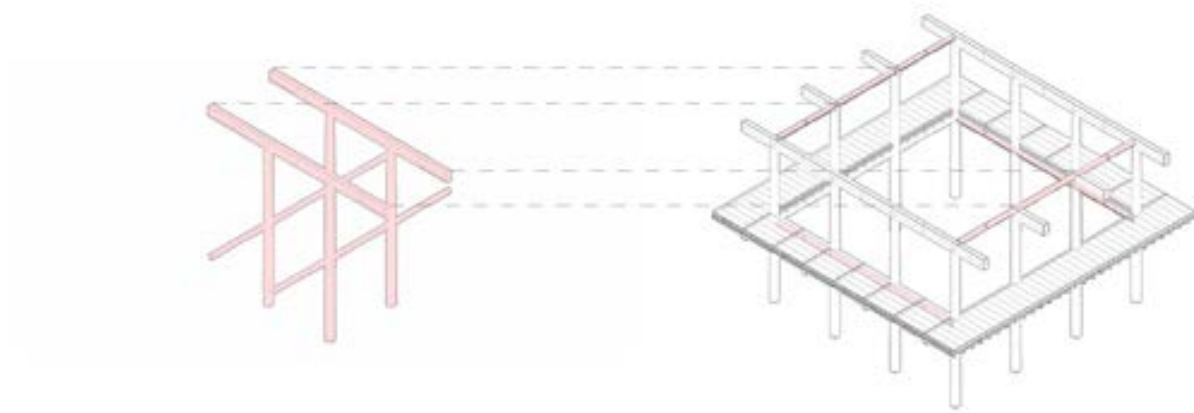
The sequence shows adjustments to the roof, floor openings, structural arrangement and enclosure around the central ritual space. A cable-driven parallel robot (CDPR) is incorporated into the atrium, with winches integrated into the surrounding seating. These changes explore how the equipment and its movement can be accommodated alongside occupation and religious practice.



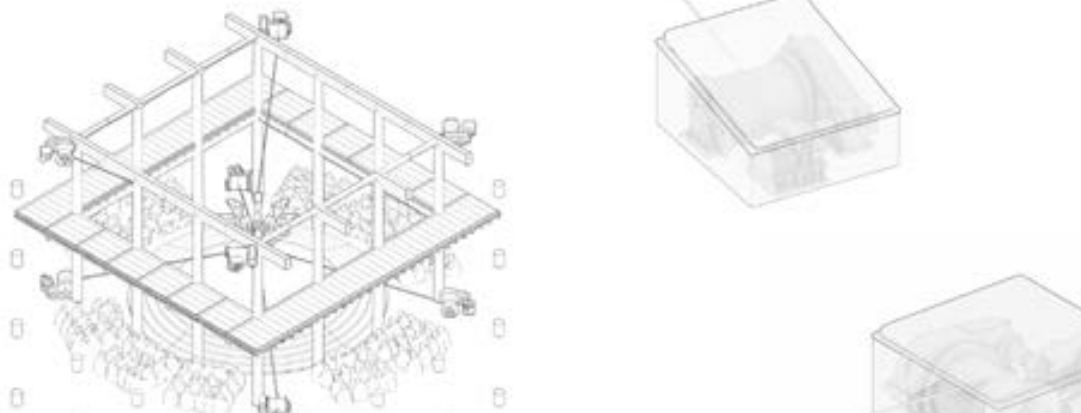
D' DOUBLE FLOOR WITH 9 x 9 M CENTRAL VOID



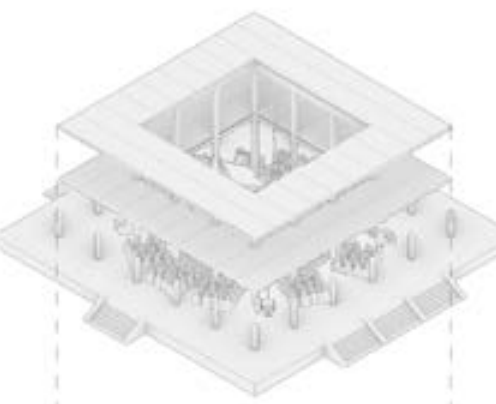
E ATRIUM WITH INTEGRATED CDPR SYSTEM E' CDPR SYSTEM CONFIGURATION



D'' STRUCTURE REMOVED D DOUBLE FLOOR WITH 9 x 9 M CENTRAL ATRIUM



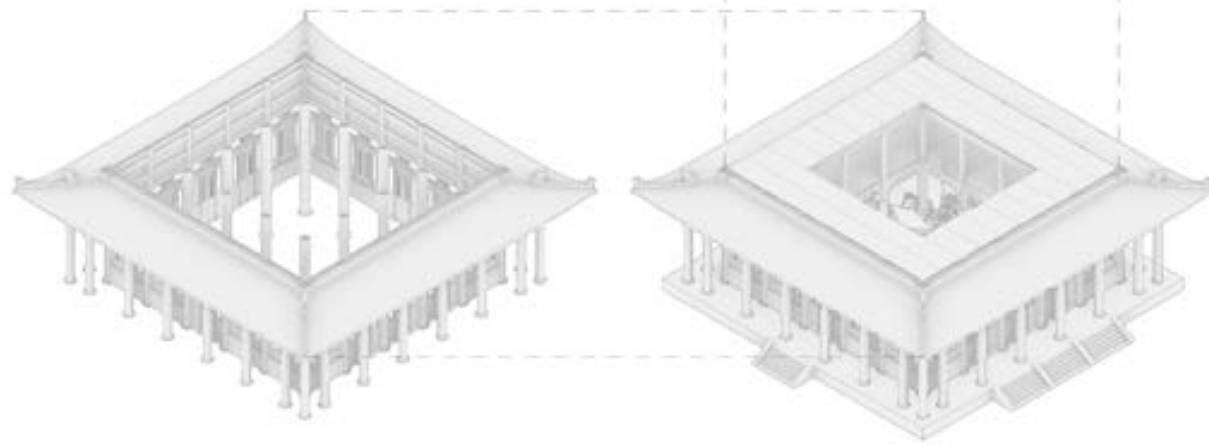
F SAND MANDALA RITUAL OPERATION G WINCH INTEGRATION WITHIN TRADITIONAL SEATING



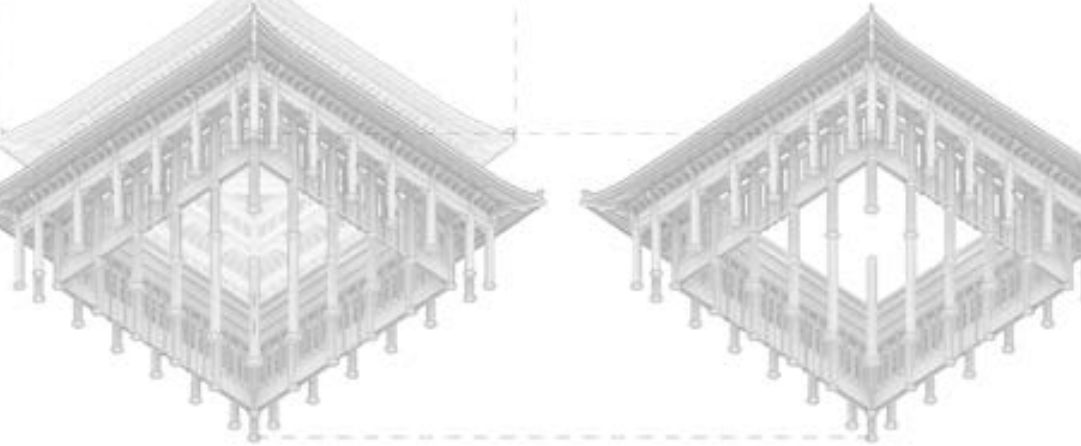
H CORE STRUCTURAL SYSTEM FOR CDPR OPERATION



I' FIRST FLOOR CLADDING



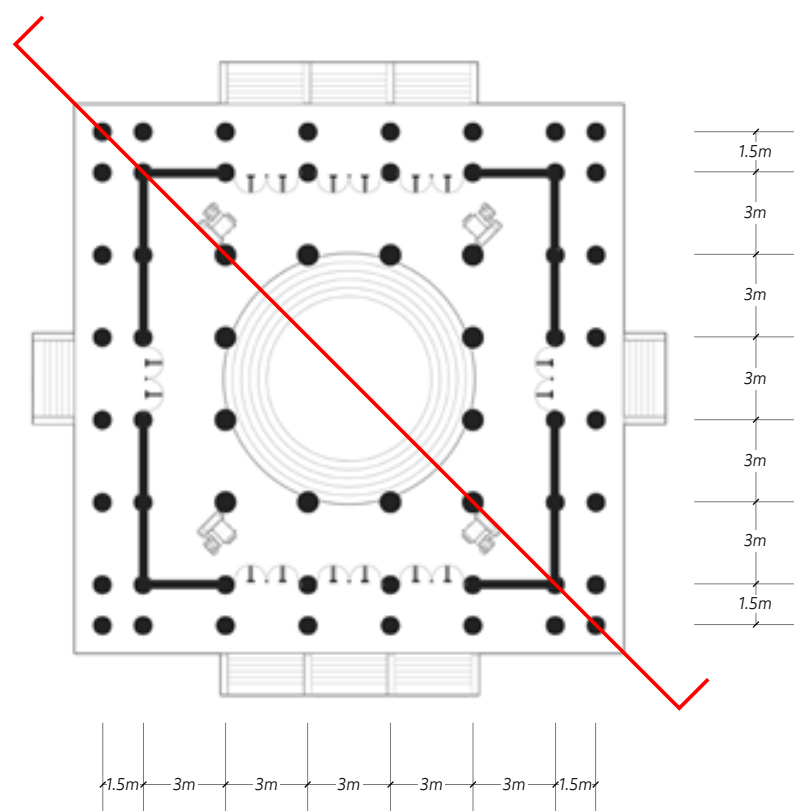
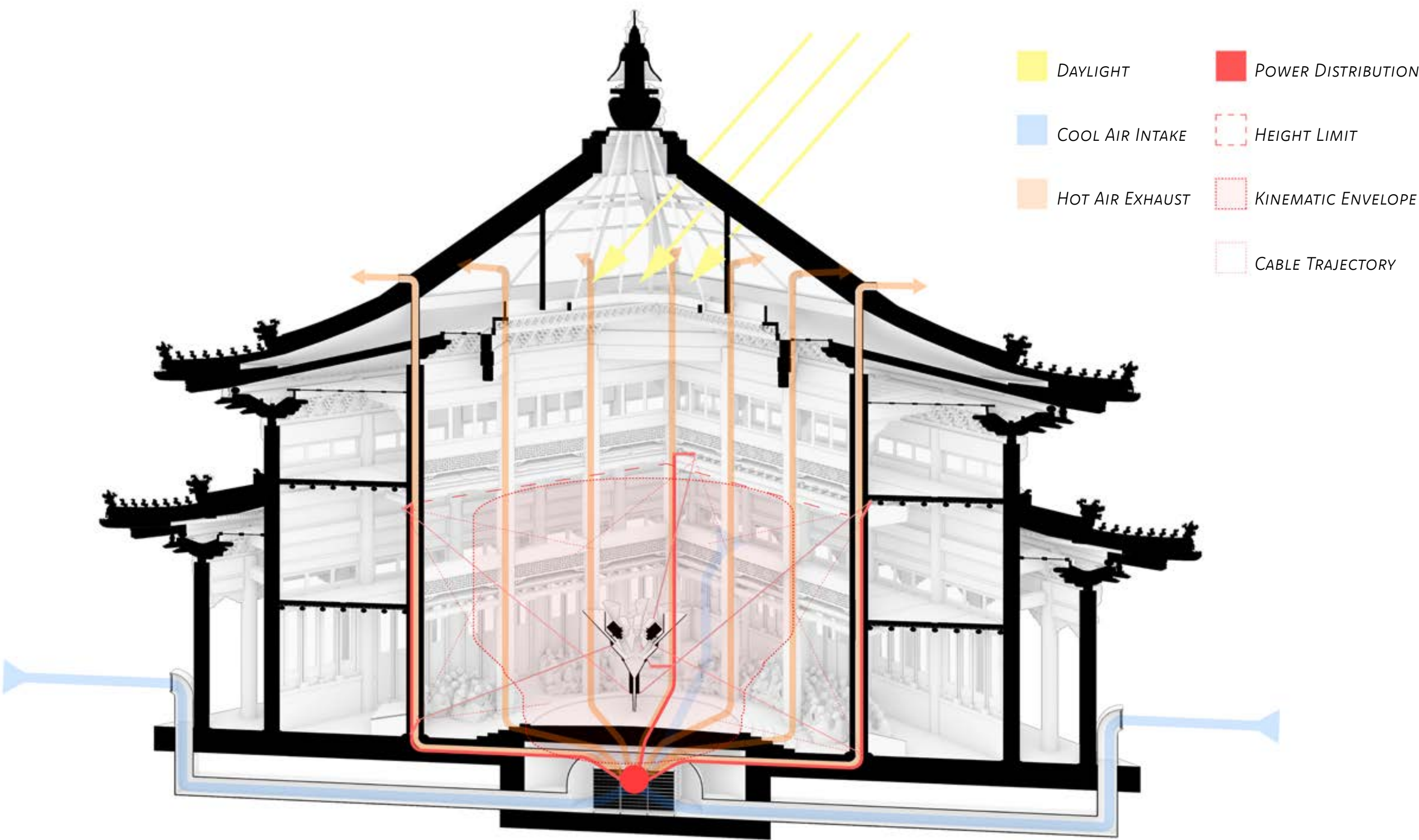
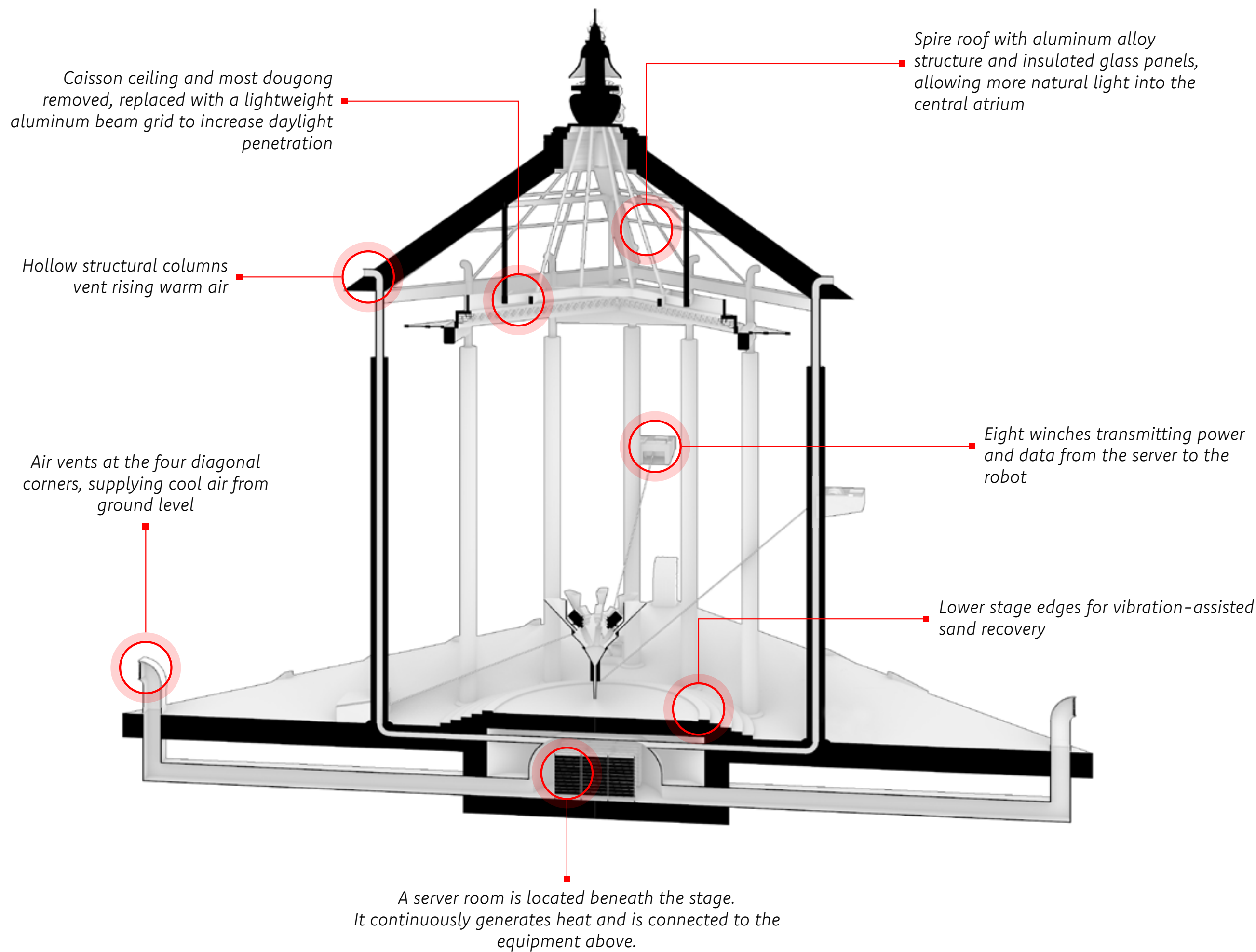
H' GROUND FLOOR CLADDING H'' INTEGRATED STRUCTURE AND CLADDING



I'' COMBINED CLADDING OF THE FIRST TWO FLOORS I GROUND FLOOR CLADDING

The sections develop the relationship between the robotic system and the building's structure, enclosure and services. Roof alterations are intended to admit daylight into the central atrium, while the proposed ventilation strategy uses low-level air inlets and hollow columns to provide an exhaust route for warm air.

A server room beneath the stage supports the installation, with dedicated routes for power and data. The robotic movement envelope and cable trajectories are considered alongside the occupied space, bringing equipment requirements into the architectural design.



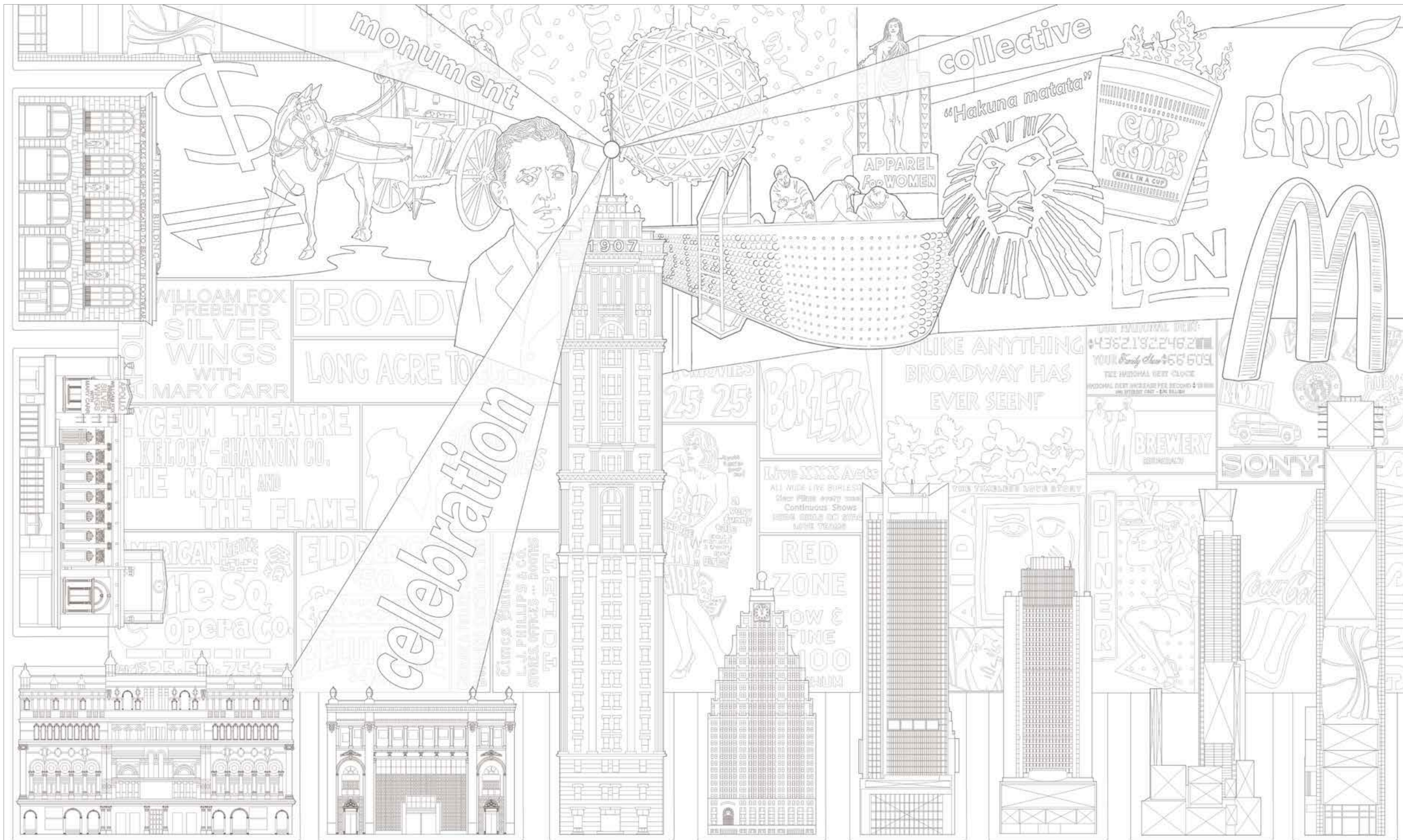
Together, these studies develop the proposal from a tested making process towards a building-scale system, connecting ritual use with the spaces and infrastructure that would support it.

3. LIGHT UP NEW YORK FOR BUT A MOMENT

THE URBAN FAÇADE
MEDIA AS ARCHITECTURE
AN ANNUAL URBAN RITUAL
STRUCTURE & OCCUPATION

Light Up New York for But a Moment investigates how advertising has reshaped the architecture of Times Square. The area's emergence as a centre of theatre, commerce and electric signage established a lasting relationship between the building façade and public spectacle.

The drawing traces successive layers of signage and media across the square. It provides the starting point for a proposal that detaches advertising from existing buildings and gives it a temporary architectural form of its own.



Billboard Evolution

The four studies trace the changing relationship between buildings and advertising in Times Square. As signage expands in scale and technical capacity, the façade increasingly becomes a support for an independent media system.



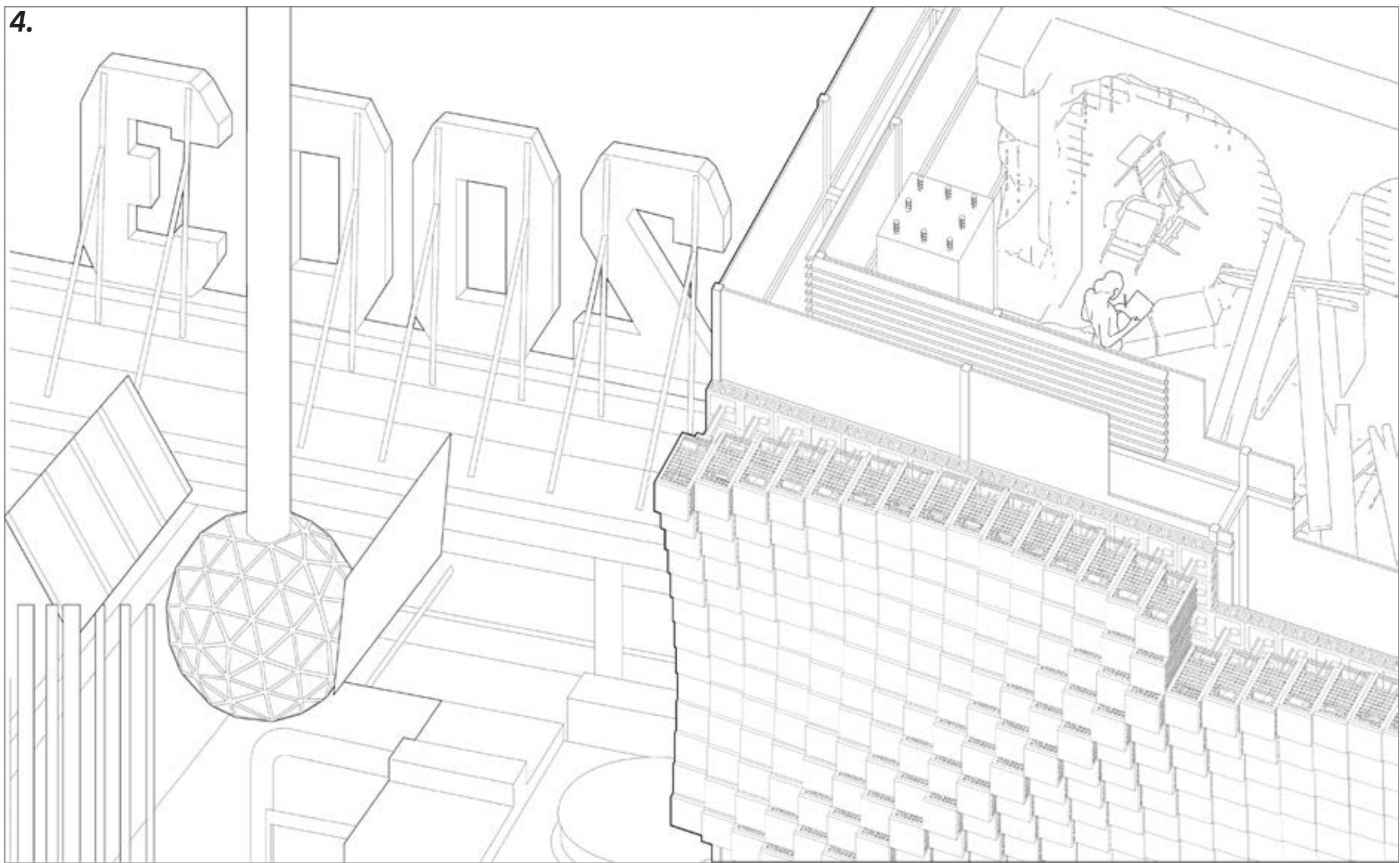
1.



2.



3.



4.

1. Theatre Culture and Illuminated Signage

Early illuminated signs announced theatres and businesses while remaining closely tied to the scale and composition of their host buildings.

2. Blackout, Decline and Neon

This study follows the interruption of illumination during wartime and the prominence of neon within the district's changing commercial and social landscape.

3. Disneyfication and the LED Turn

Redevelopment and entertainment-led investment reshaped the square, while LED displays extended the scale and intensity of advertising across its façades.

4. Digital Media and Façade Loss

Large digital displays form increasingly continuous media envelopes, obscuring the buildings behind them and shifting attention from architectural frontage to broadcast surface.

Media Breaks Free

The proposal detaches billboards from the buildings of Times Square and gathers them into a single media sphere. Existing façades are revealed, while the displaced advertising acquires a structure and interior of its own.

This shift establishes the project’s central architectural question: what happens when a medium designed to be viewed from outside becomes a space that people can enter?

1. Times Square Today

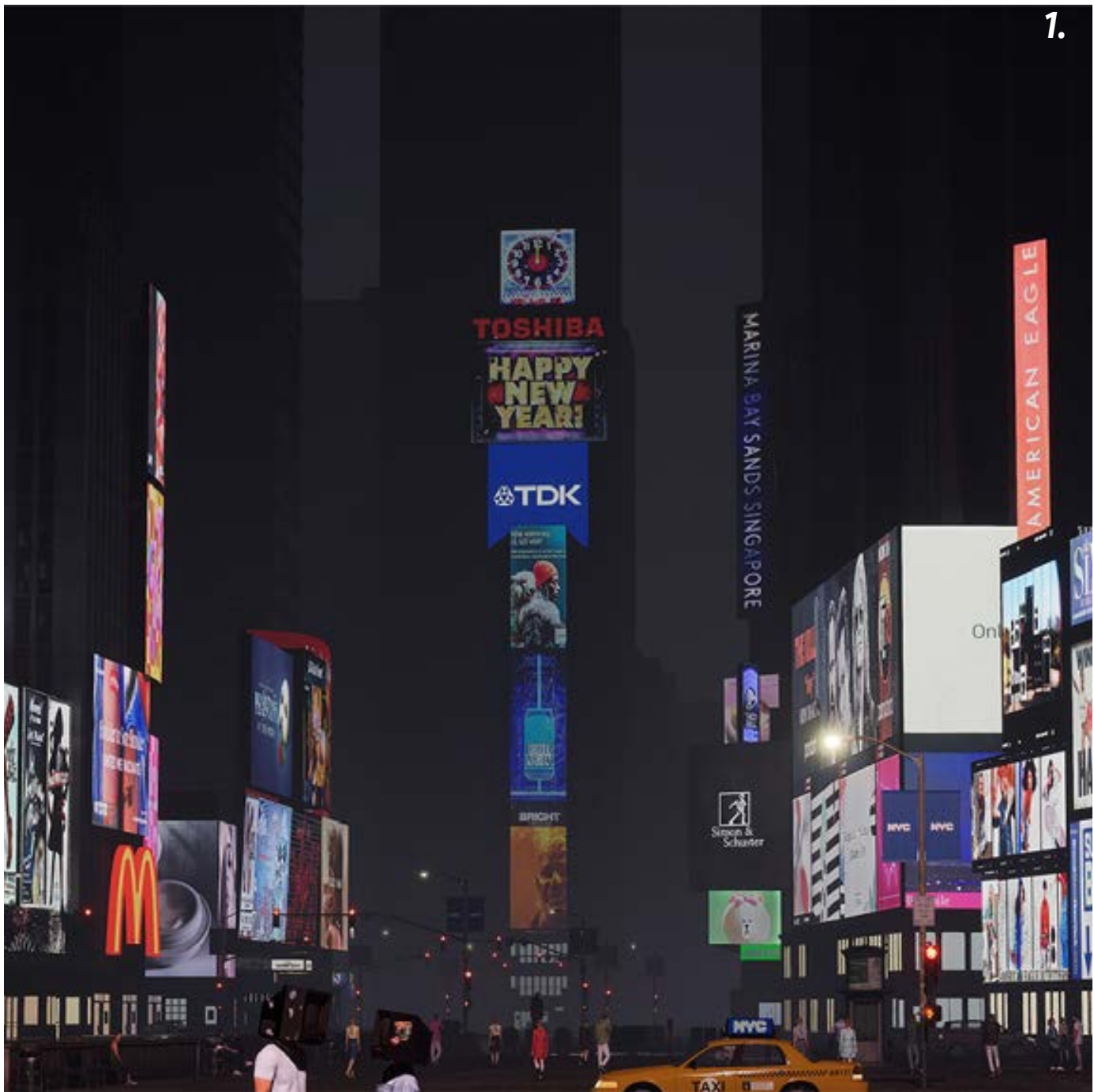
Billboard structures and digital displays conceal much of the architecture facing the square.

2. Media Sphere and Revealed Façade

The proposal concentrates these surfaces into a single object, exposing the buildings that previously supported them.

3. Architectural Possibility

The supporting frames, circulation routes and depth behind the screens become the basis for an inhabitable interior.

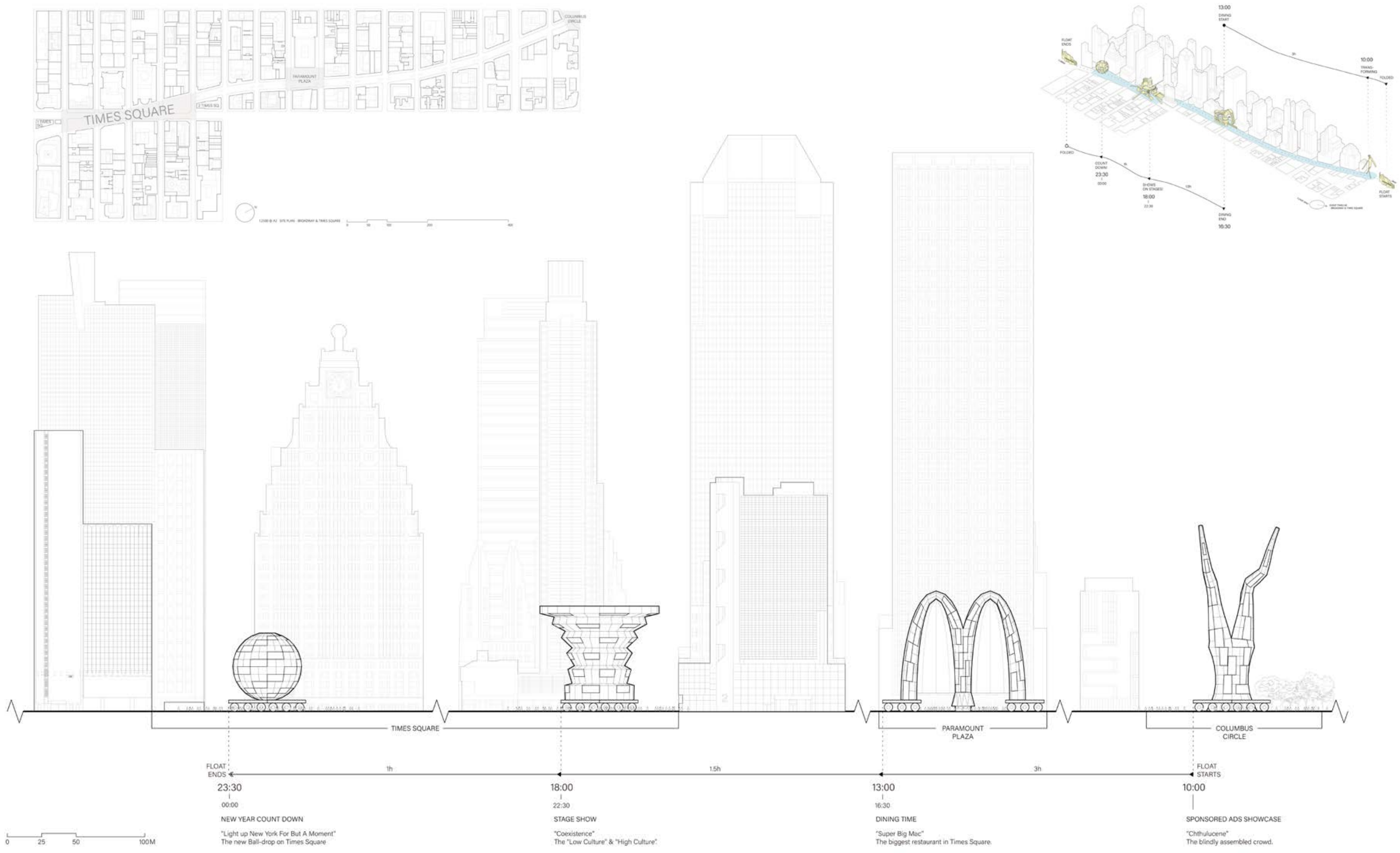


AN ANNUAL URBAN RITUAL

CAD, RHINO, ADOBE CREATIVE CLOUD
UNREAL ENGINE

The project reimagines the New Year’s Eve Ball Drop as the culmination of a temporary urban ritual. For most of the year, advertising is removed from the surrounding façades. During the annual celebration, it returns as a transformable parade float travelling along Broadway towards One Times Square.

The structure moves through a sequence of commercial and performative identities before becoming the New Year’s Eve Ball. By concentrating advertising into a scarce annual event, the proposal explores how the city might recover its façades while intensifying the spectacle of their temporary replacement.



1. **Chthulucene**

At 10 AM, the structure enters the city as the Chthulucene: strange, shifting and deliberately unfamiliar. Tourists and New Yorkers gather behind it. Advertising no longer waits to be seen. The city follows it.

2. **Big-Mac**

By noon, the screens turn gold and the structure reforms into the Golden Arches. In this imagined scenario, McDonald's concentrates its advertising expenditure into a monumental moving restaurant. Customers, vendors and commerce gather around it. For one annual event, a brand becomes architecture at the scale of the city.

3. **Coexistence**

At 6 PM, the structure opens into an urban stage. Celebrities occupy its illuminated exterior, while the Naked Cowboy performs among the screens, cables and machinery. Global entertainment and local street performance briefly share the same architectural body.

4. **Light Up New York for But a Moment**

At midnight, the earlier identities disappear. The structure reforms into a ball and moves towards One Times Square. The crowd below and audiences watching remotely turn towards the same object.

Bigger than ever. Brighter than ever.

5 — 4 — 3 — 2 — 1.

Happy New Year.

Same as usual.

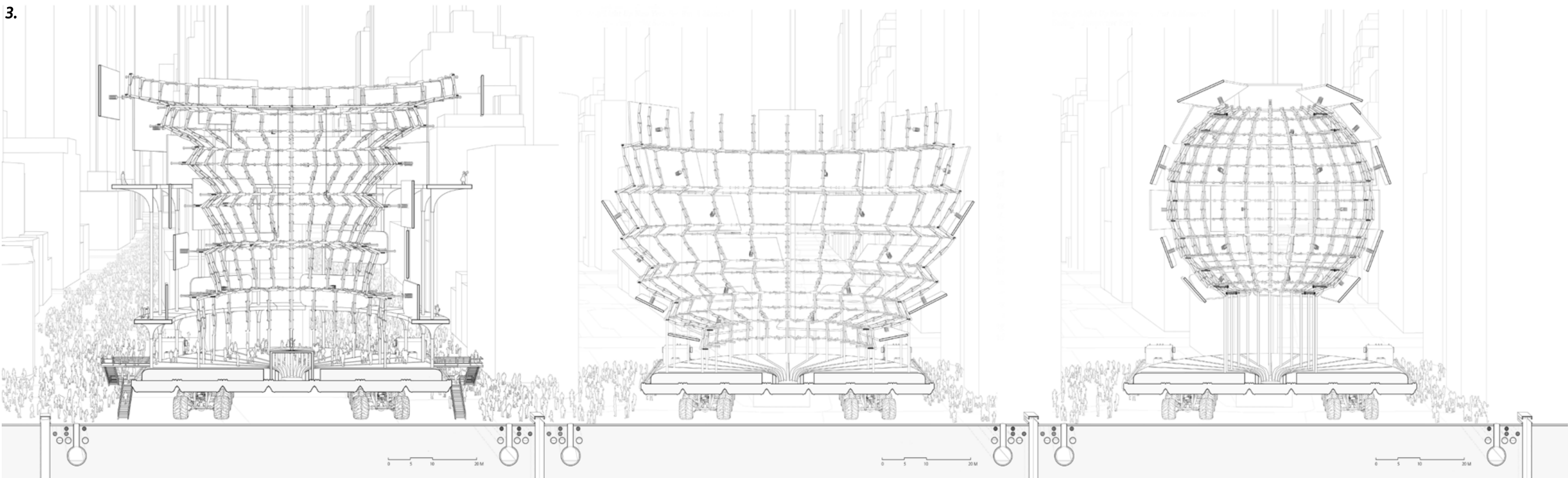
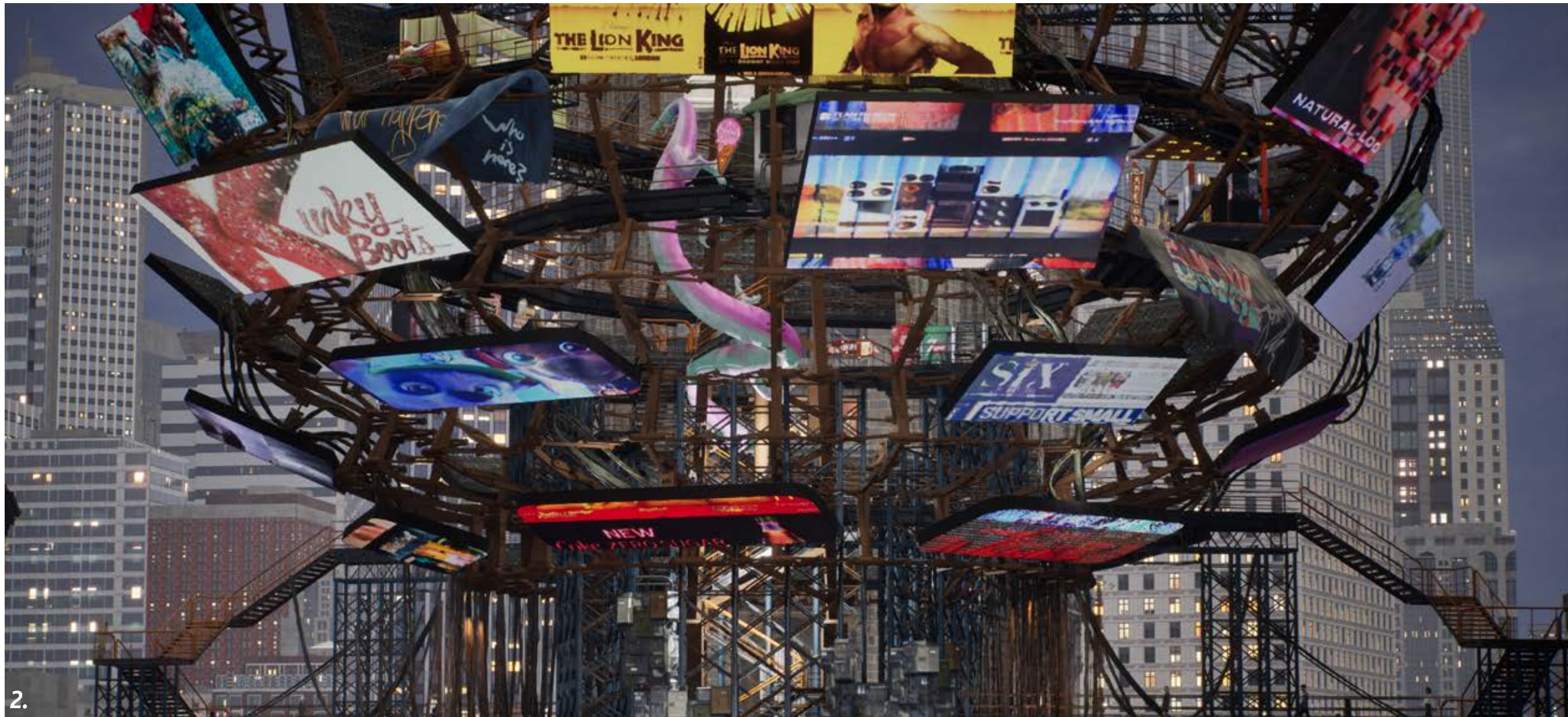
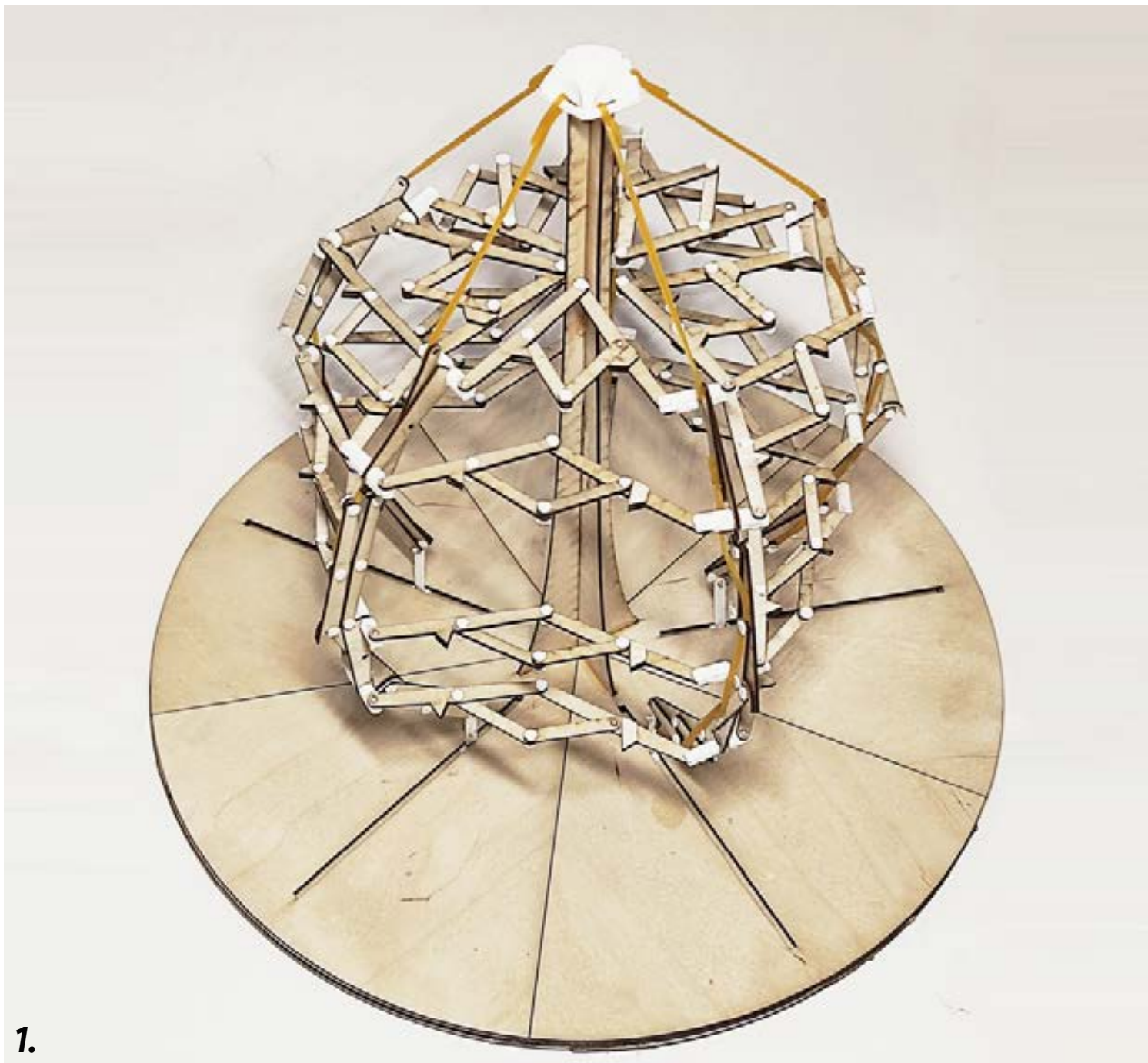


Final film – 8 mins



STRUCTURE & OCCUPATION

CAD, RHINO, ADOBE CREATIVE CLOUD
PHYSICAL MODEL, UNREAL ENGINE



Structure, Movement, Form

The media sphere is developed as a transformable assembly of supporting frames, screens and circulation spaces. The drawings and physical model investigate how its geometry changes between the configurations shown in the parade sequence.

The structural system becomes an architectural framework, shaping both the changing exterior and the spaces within it.

1. Kinetic Prototype

A physical model tests the assembly's movement between configurations.

2. Media Sphere

The visualisation explores the relationship between the screens, supporting structure and occupied interior.

3. Structural Transformation Section

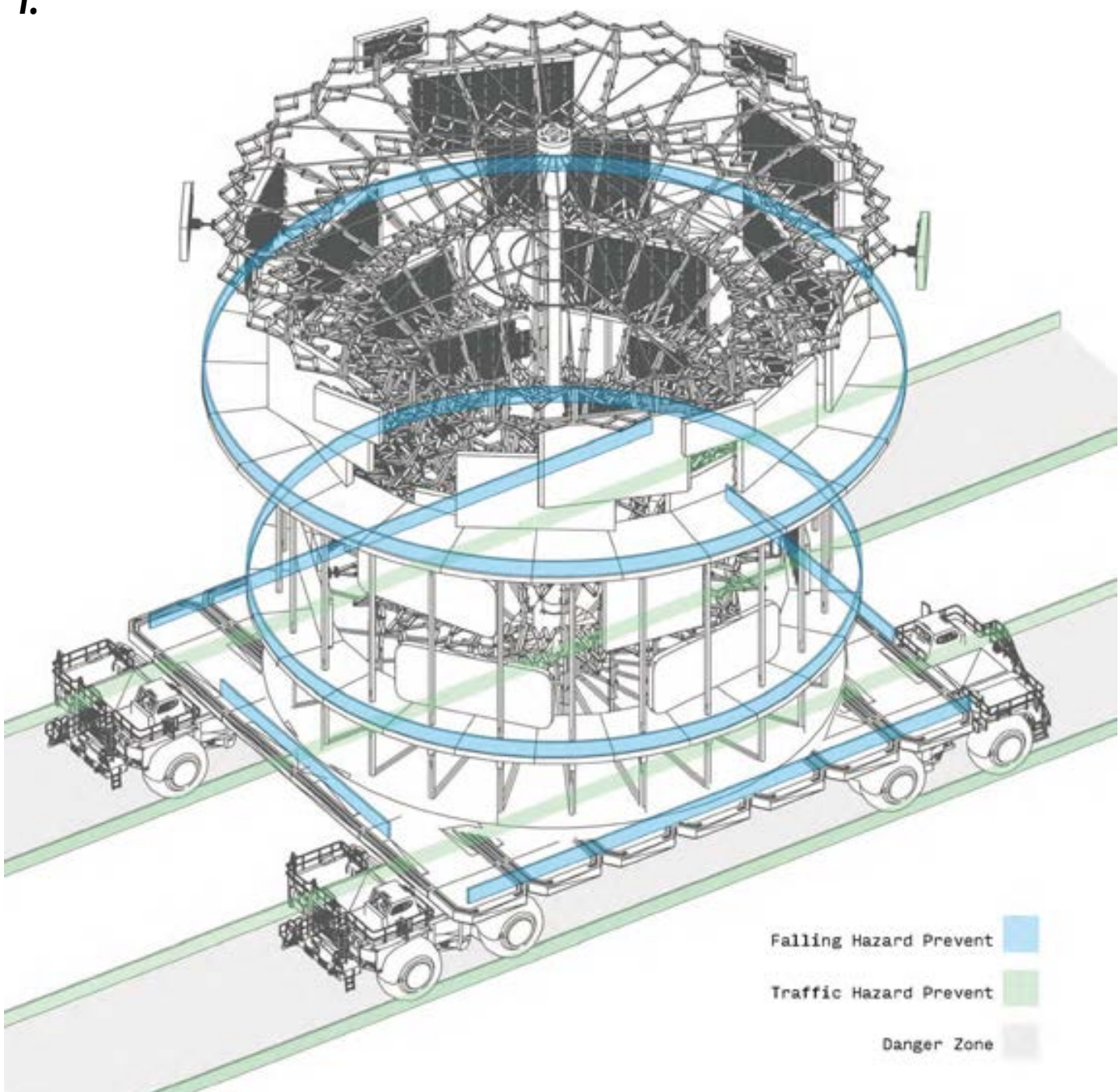
The sectional sequence follows the changing geometry of the assembly and its relationship to the street.

The Billboard Becomes Architecture

The proposal brings occupation into the space behind advertising. Screens form the enclosure, supporting frames organise the interior, and platforms and circulation routes allow visitors to move through the media structure.

Separating advertising from its host buildings produces two spatial outcomes: existing façades are revealed, and the displaced billboard becomes an inhabitable environment.

1.



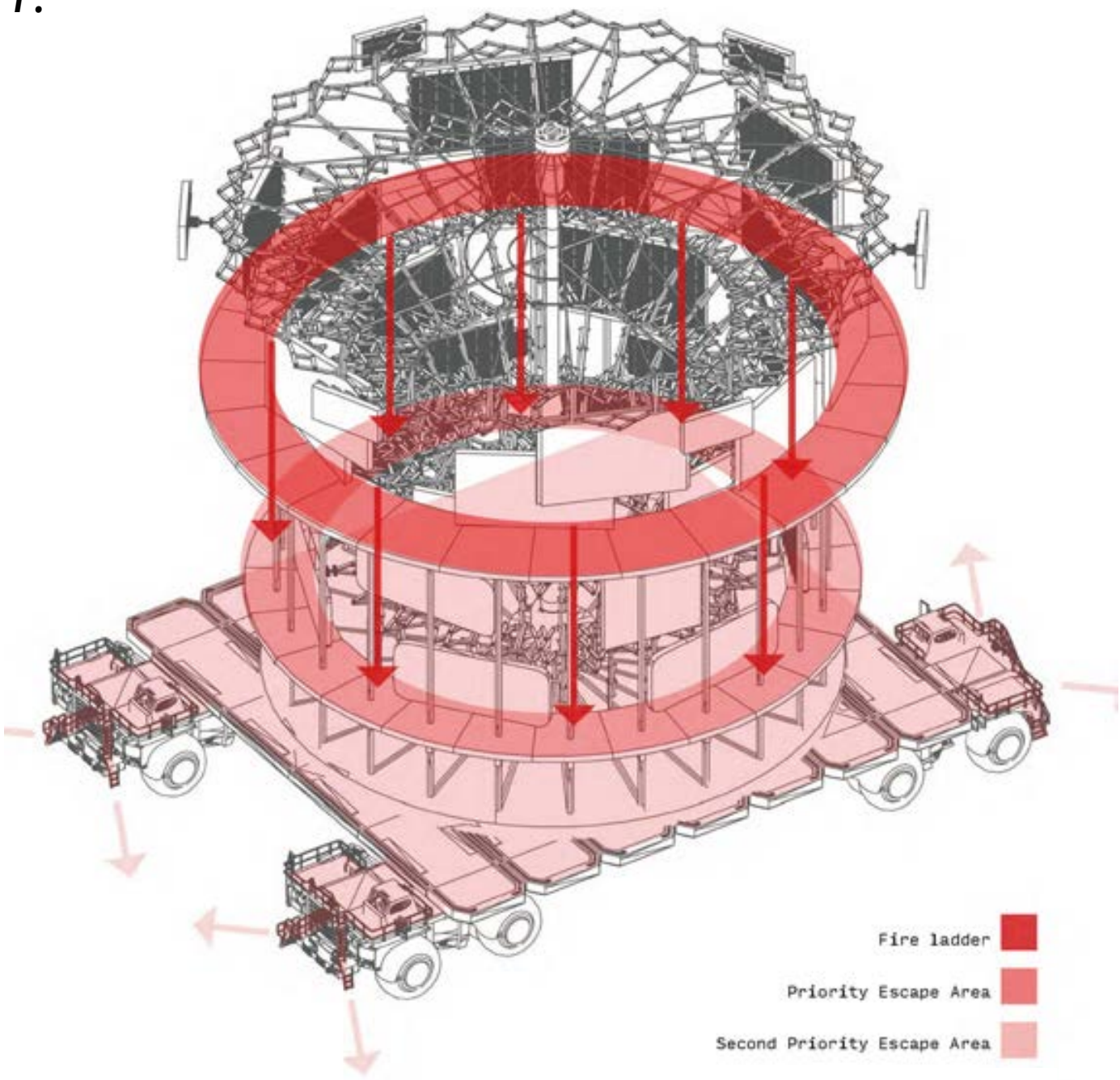
1. Operational Safety & Emergency Escape

These diagrams explore preliminary safety considerations for a mobile structure in a public street. They identify proposed exclusion zones around the float and indicative escape routes from its upper levels. The studies establish issues for further development as the structure, occupancy and operating sequence are resolved.

2. Inside the Billboard

The interior view explores occupation among the screens and supporting frames, bringing circulation, activity and the machinery of advertising into the same spatial experience.

1'.



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2.



4. SELECTED ARTISTIC PRACTICE

DRAWING, SCULPTURE & SPATIAL STUDIES

Sculpture and hand drawing have been a continuous part of my work, developing my understanding of form, proportion, material and spatial depth.

These selected studies extend from observational work to spatial constructions and continue to inform my approach to architectural composition, physical modelling and making.

